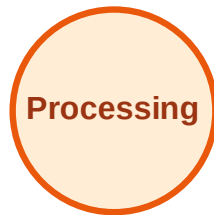


BFS Traversal

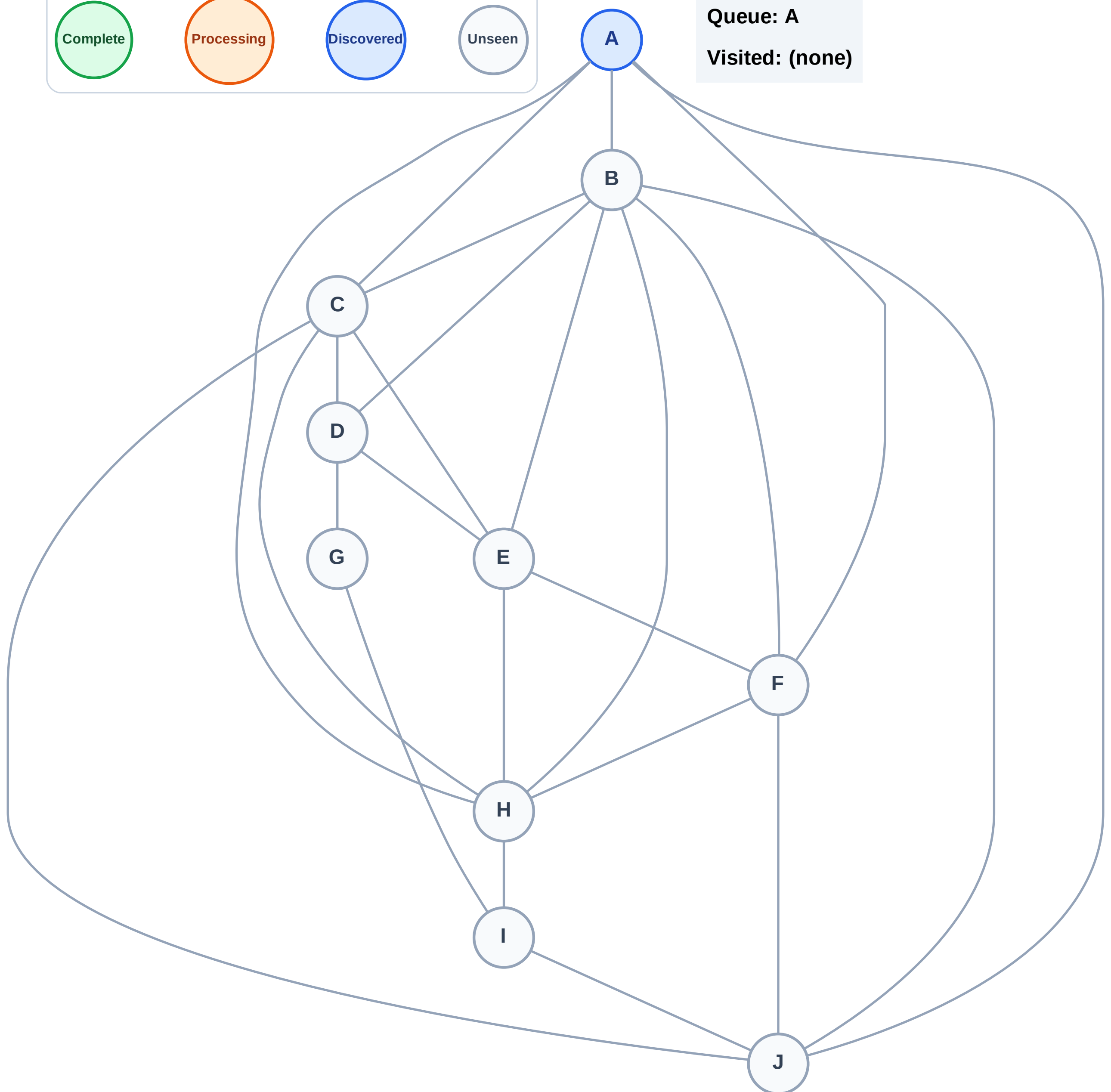
Step 1: Start at A

Legend



Queue: A

Visited: (none)



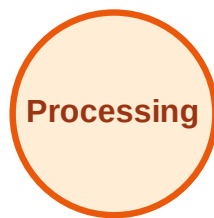
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



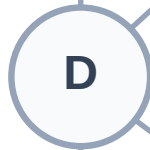
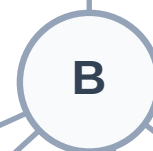
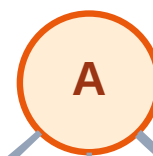
Discovered



Unseen

Queue: (empty)

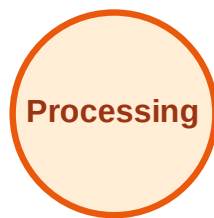
Visited: (none)



BFS Traversal

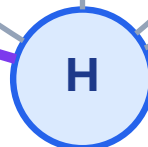
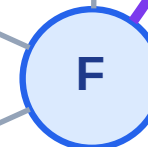
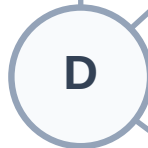
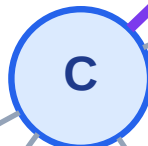
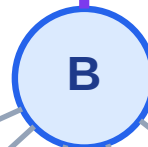
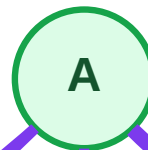
Step 3: Discover B, C, F, H, J from A

Legend



Queue: B → C → F → H → J

Visited: A



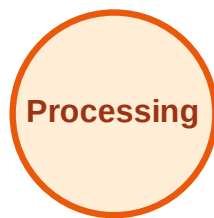
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



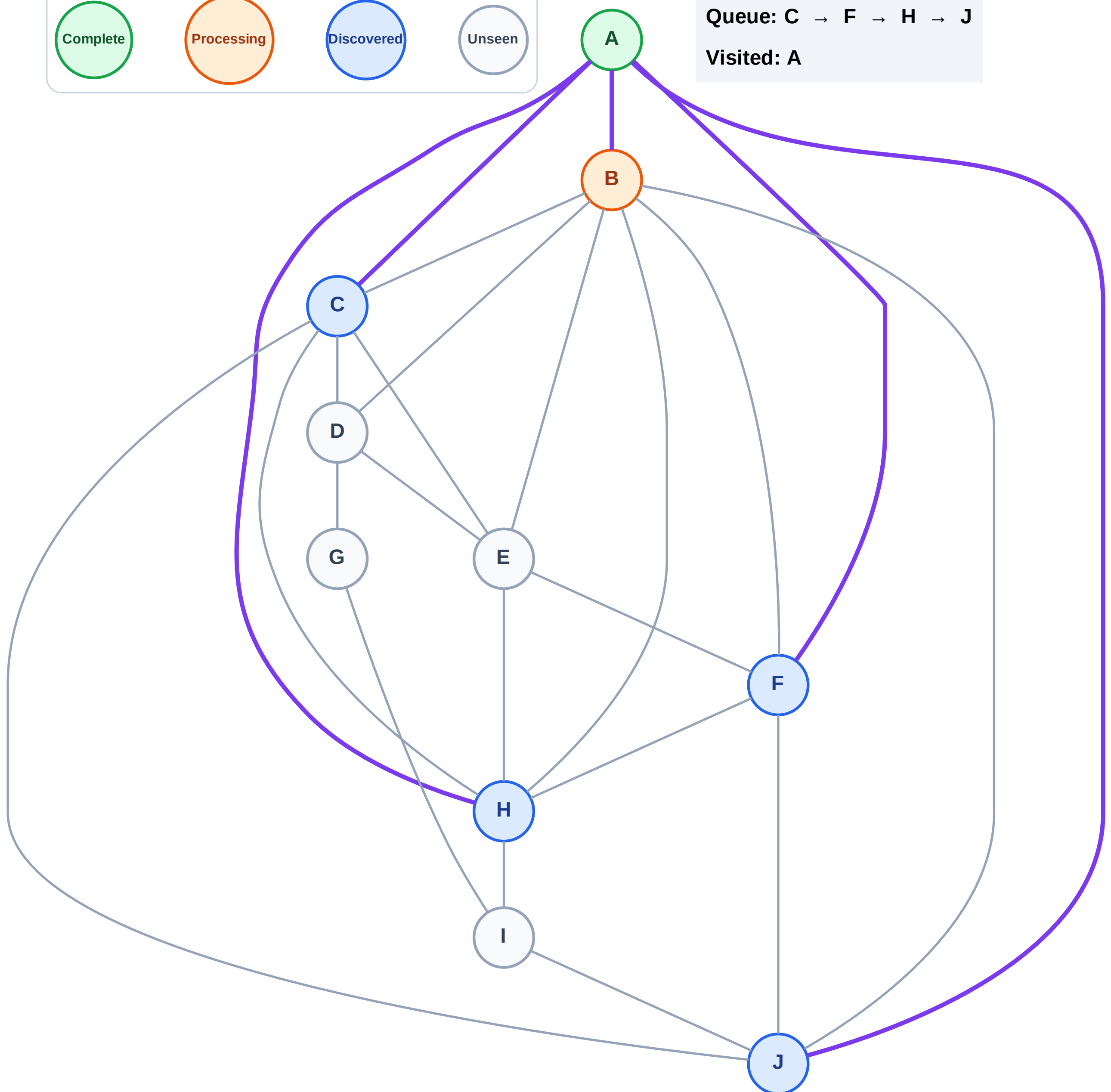
Discovered



Unseen

Queue: C → F → H → J

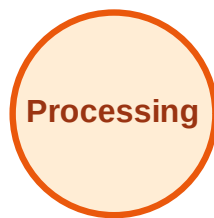
Visited: A



BFS Traversal

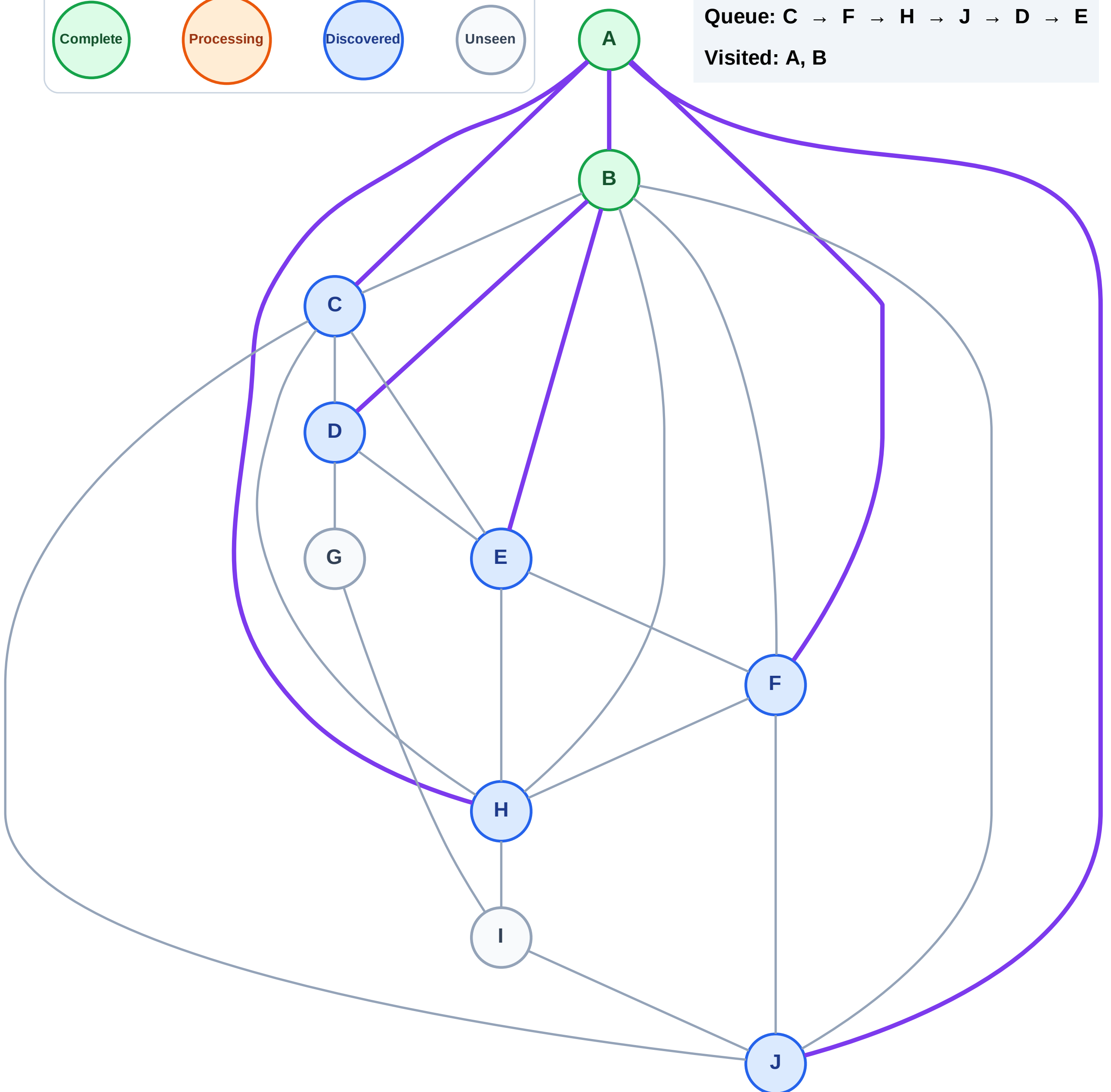
Step 5: Discover D, E from B

Legend



Queue: C → F → H → J → D → E

Visited: A, B



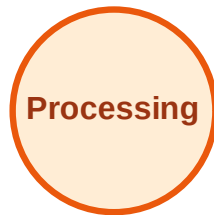
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



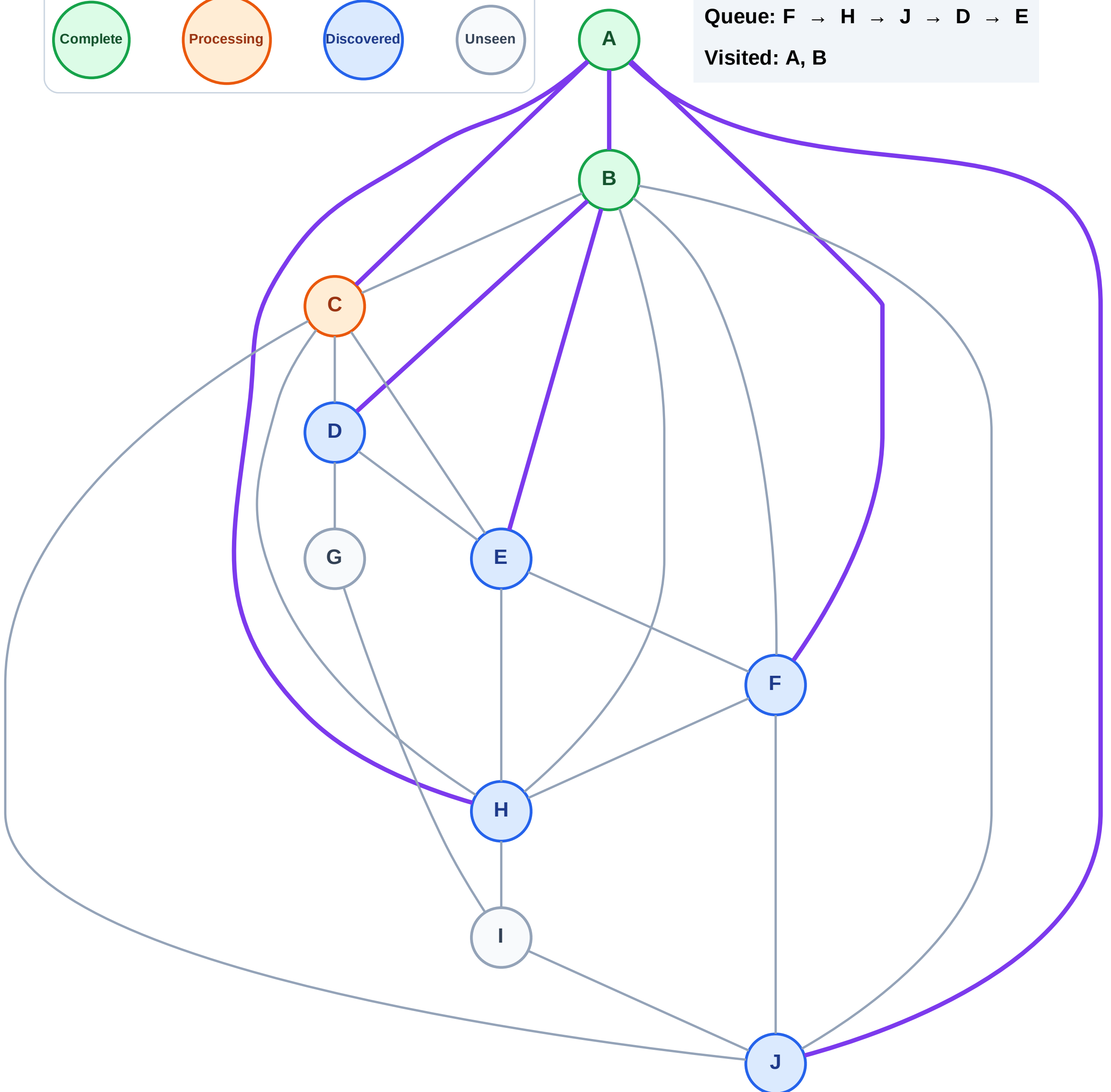
Discovered



Unseen

Queue: F → H → J → D → E

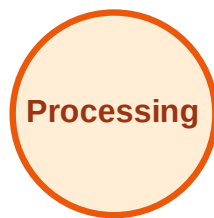
Visited: A, B



BFS Traversal

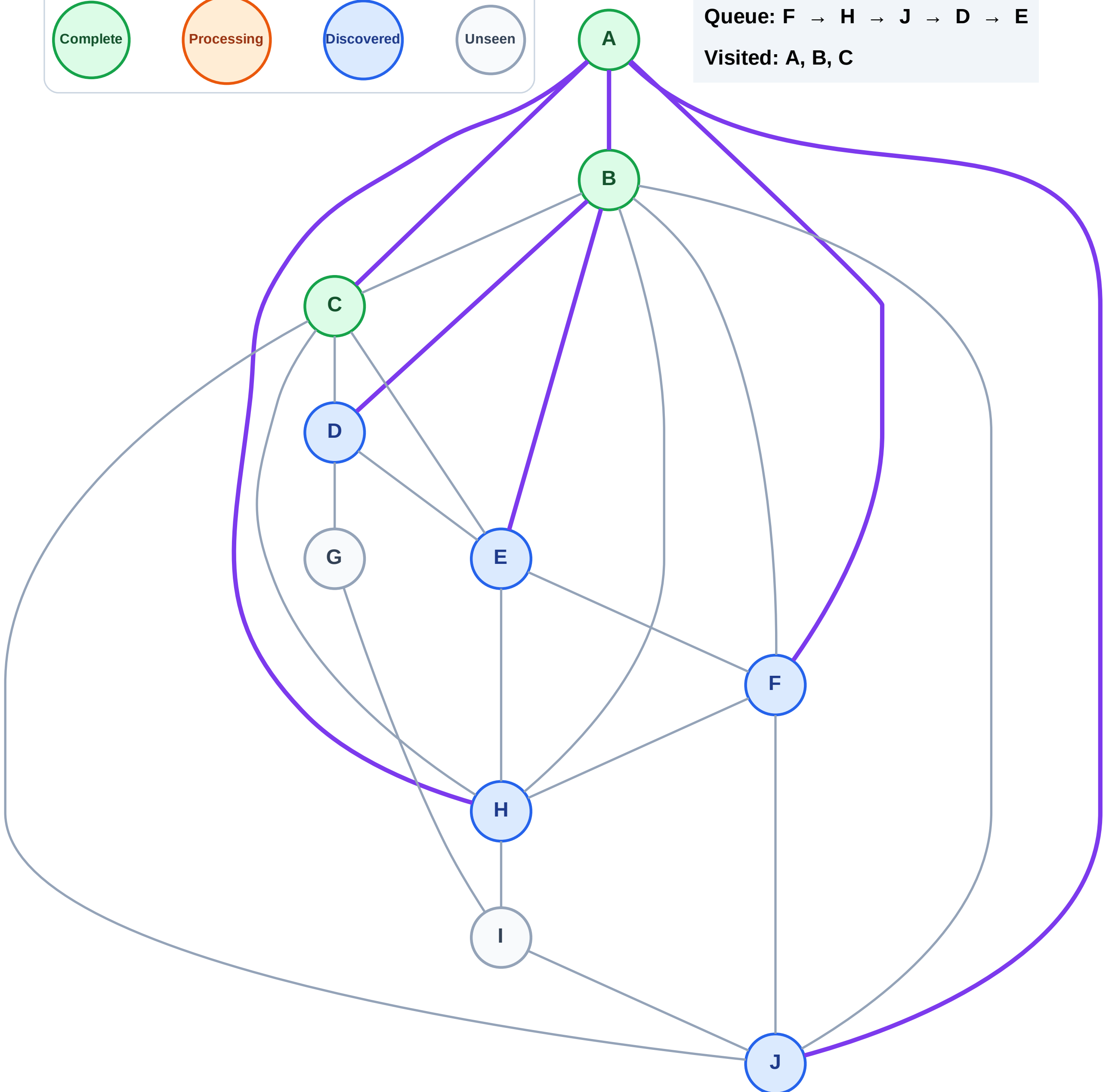
Step 7: No new neighbors from C

Legend



Queue: F → H → J → D → E

Visited: A, B, C



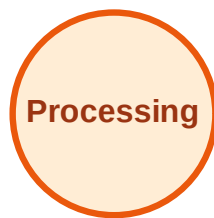
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



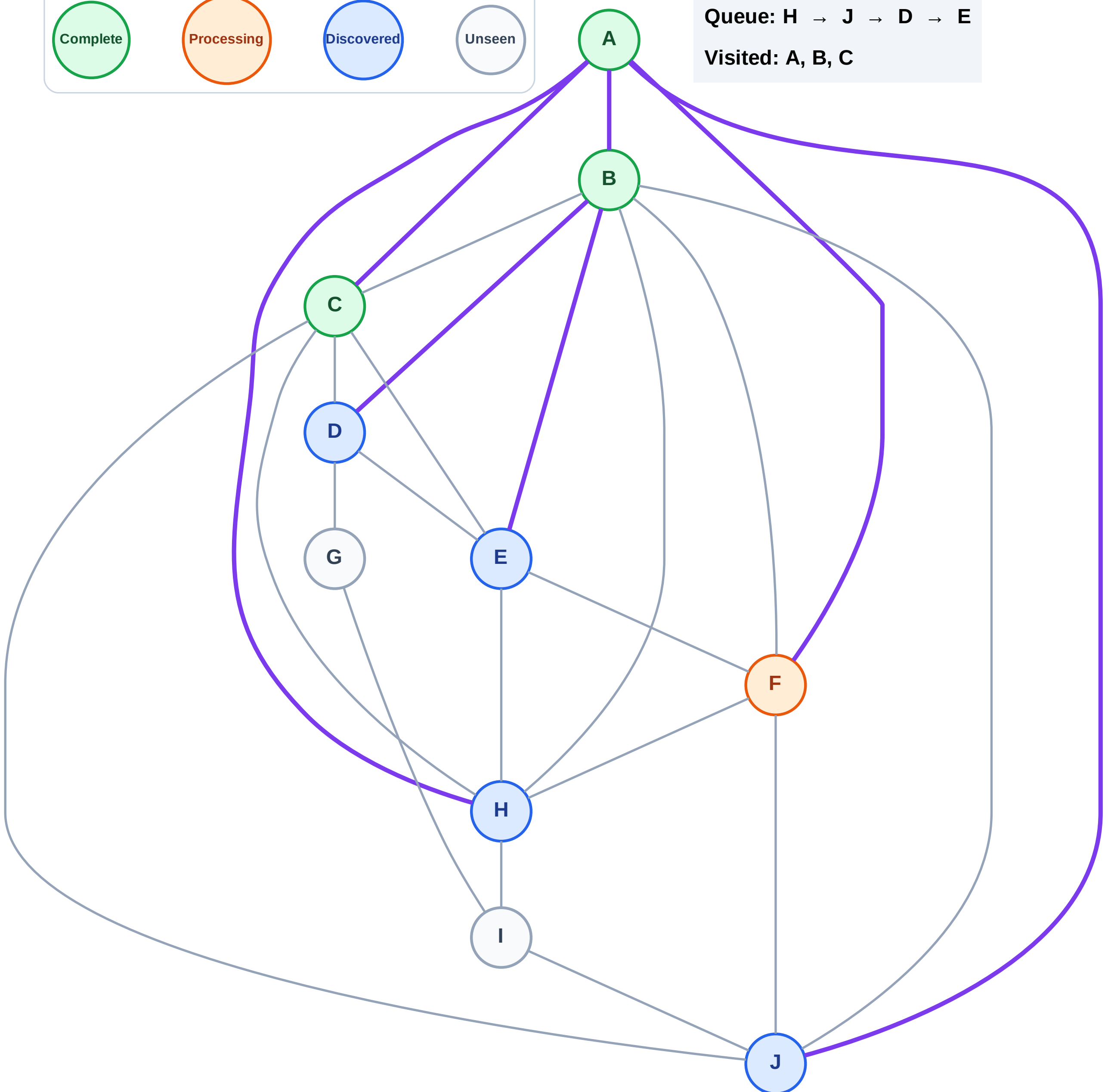
Discovered



Unseen

Queue: H → J → D → E

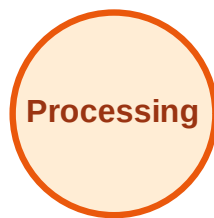
Visited: A, B, C



BFS Traversal

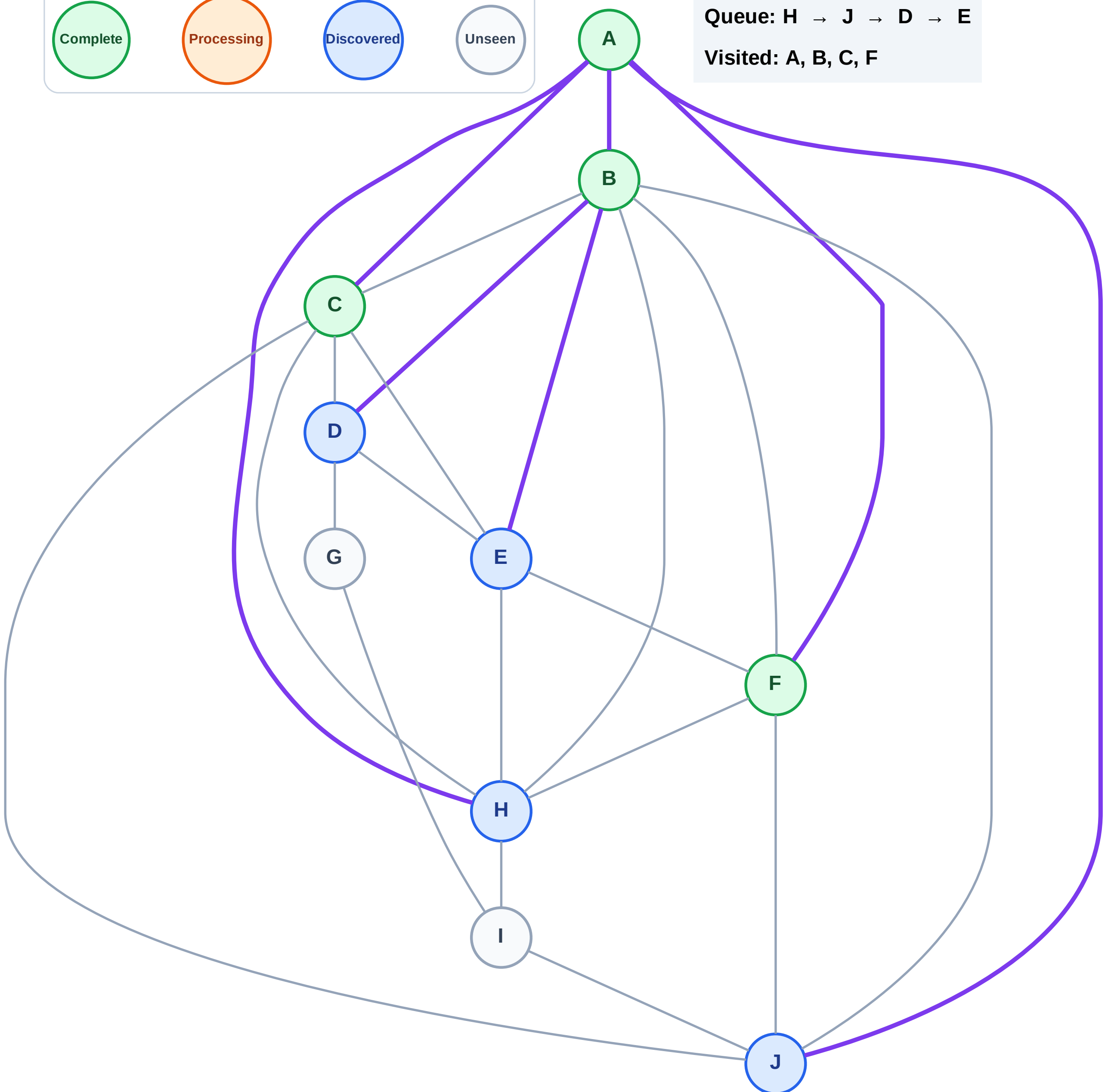
Step 9: No new neighbors from F

Legend



Queue: H → J → D → E

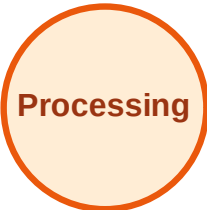
Visited: A, B, C, F



BFS Traversal

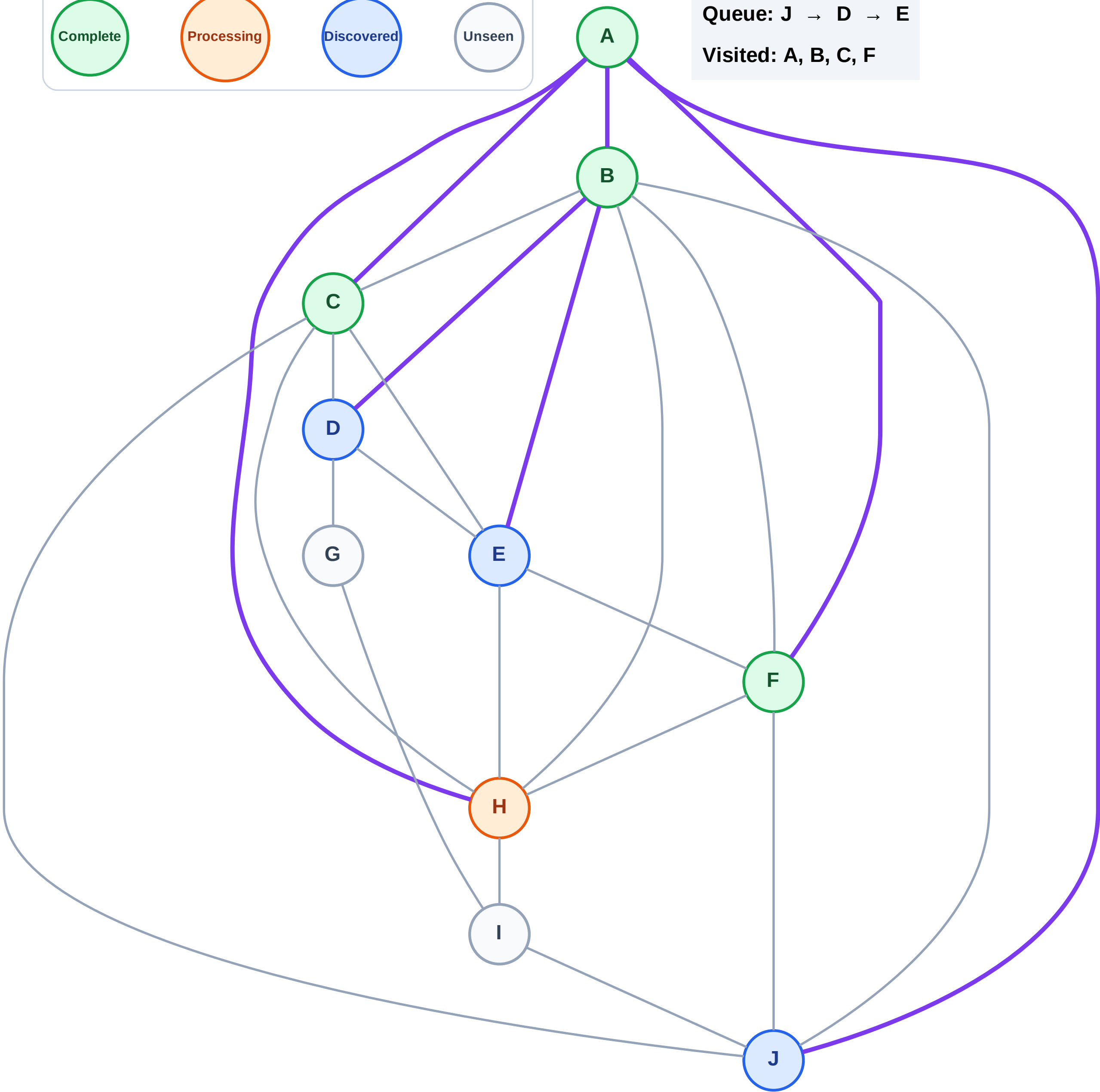
Step 10: Process node H

Legend



Queue: J → D → E

Visited: A, B, C, F



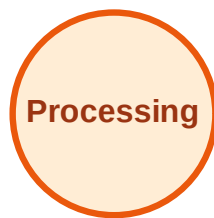
BFS Traversal

Step 11: Discover I from H

Legend



Complete



Processing



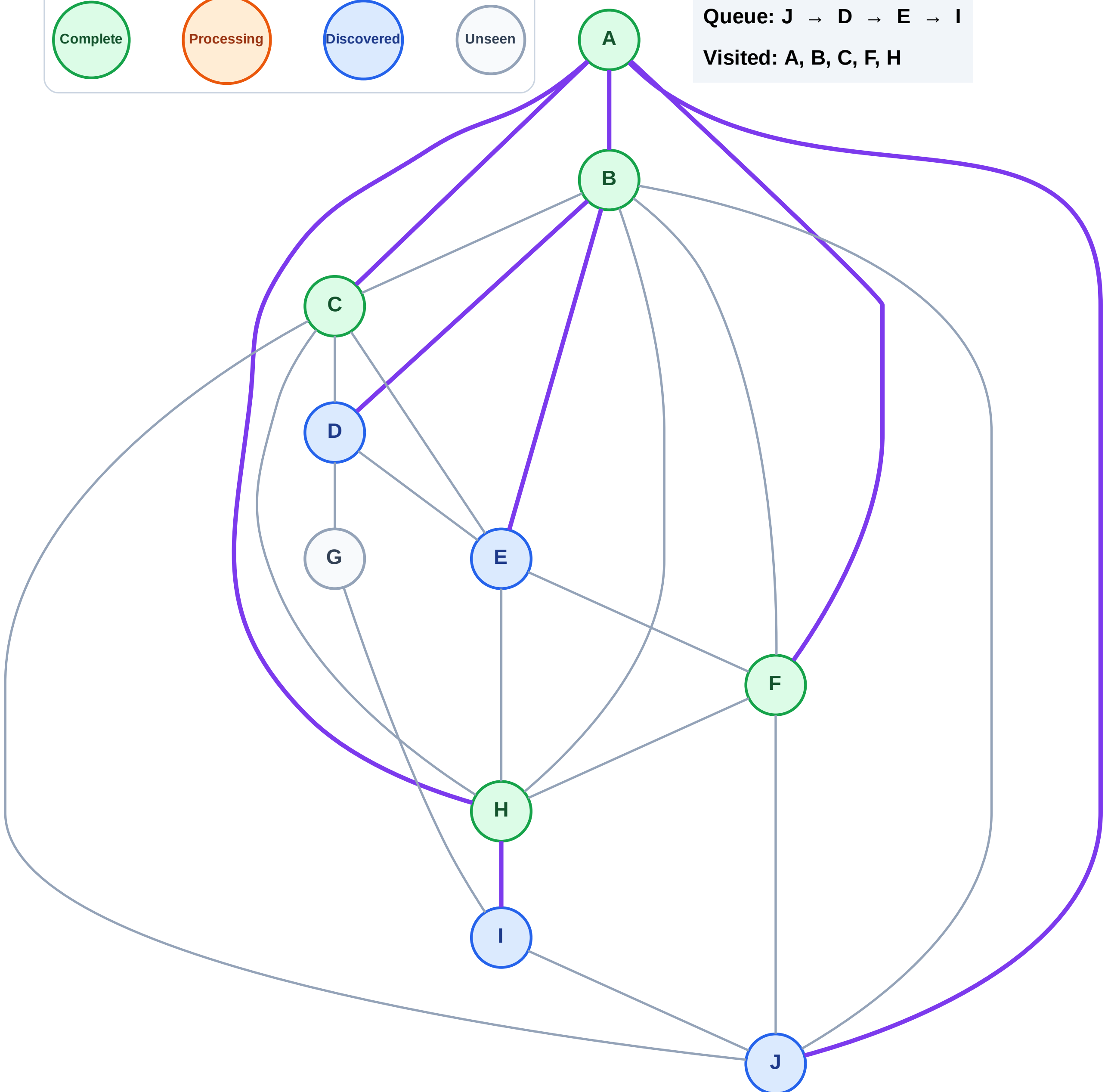
Discovered



Unseen

Queue: J → D → E → I

Visited: A, B, C, F, H



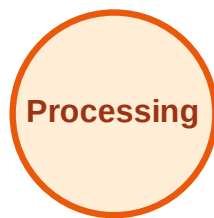
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing



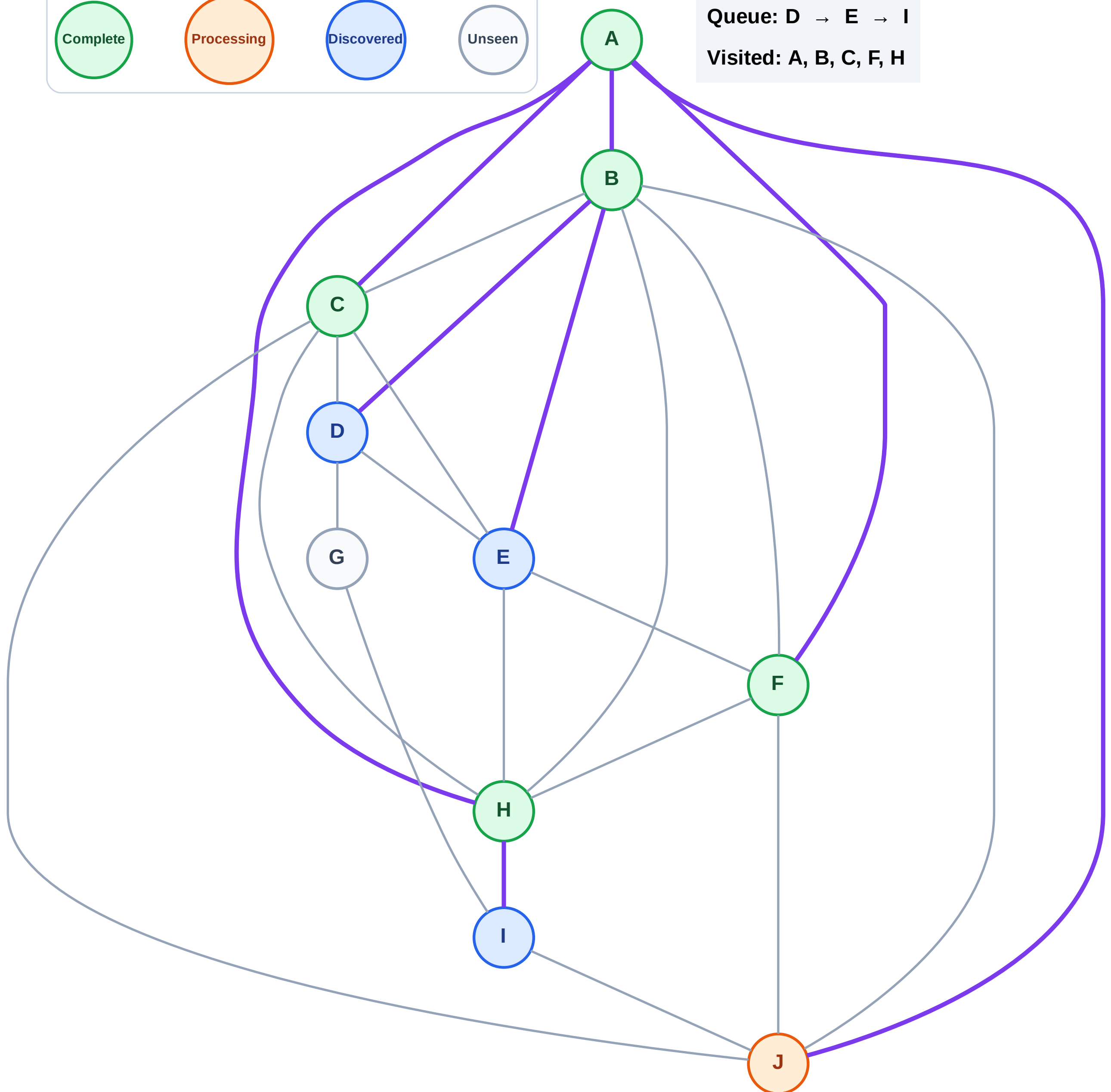
Discovered



Unseen

Queue: D → E → I

Visited: A, B, C, F, H



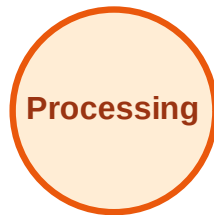
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



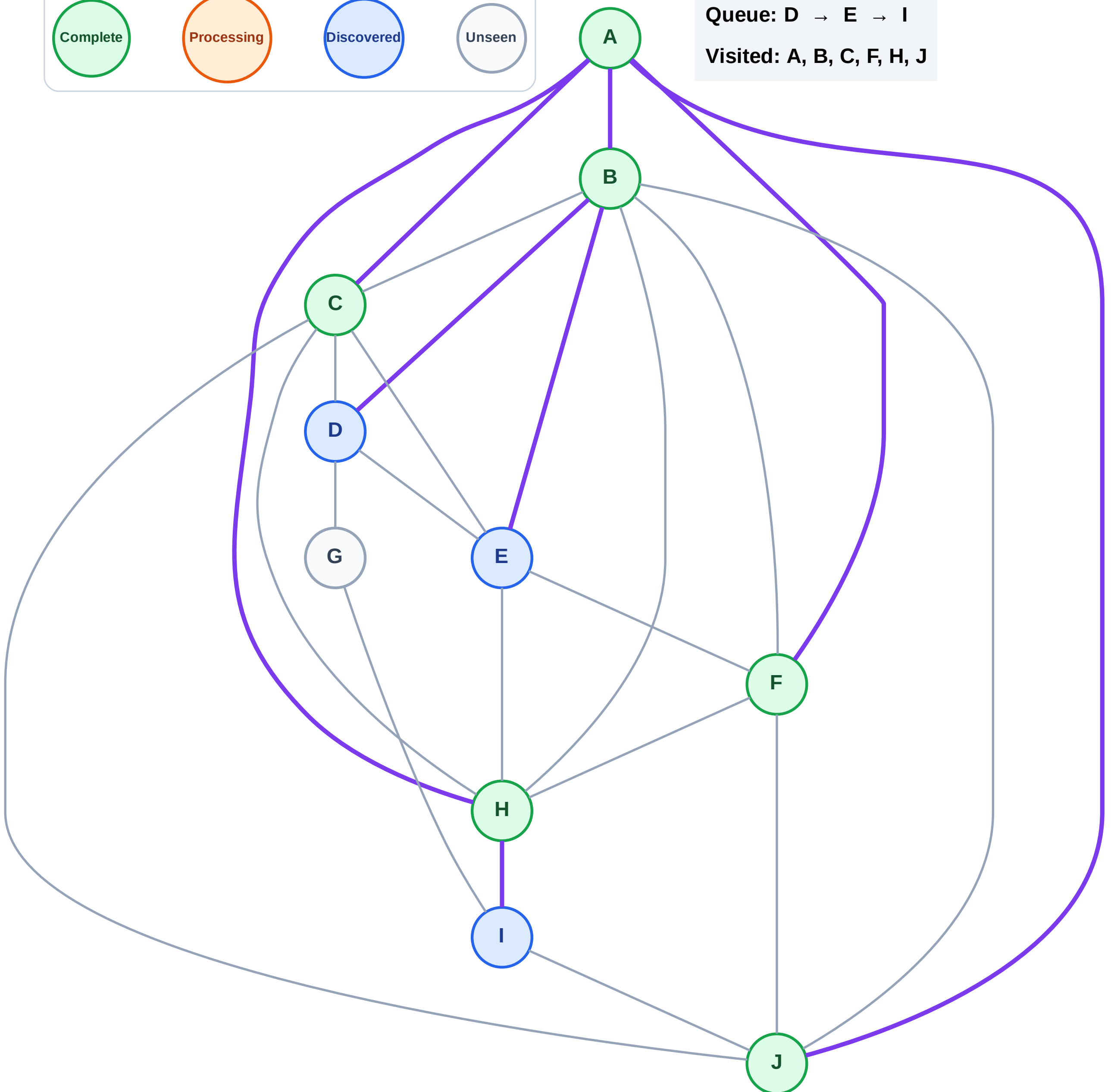
Discovered



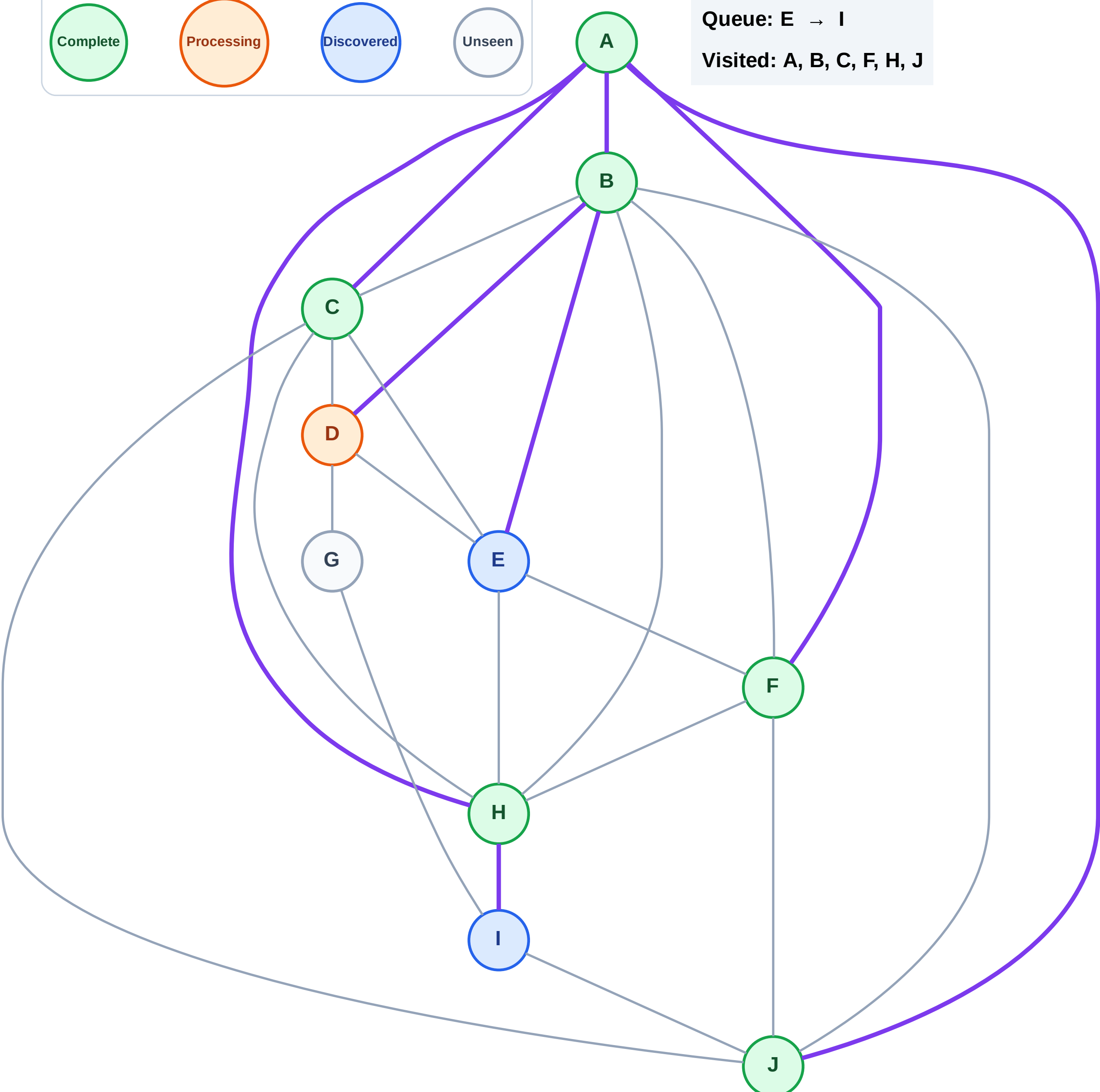
Unseen

Queue: D → E → I

Visited: A, B, C, F, H, J



Step 14: Process node D



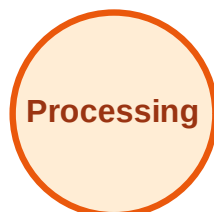
BFS Traversal

Step 15: Discover G from D

Legend



Complete



Processing



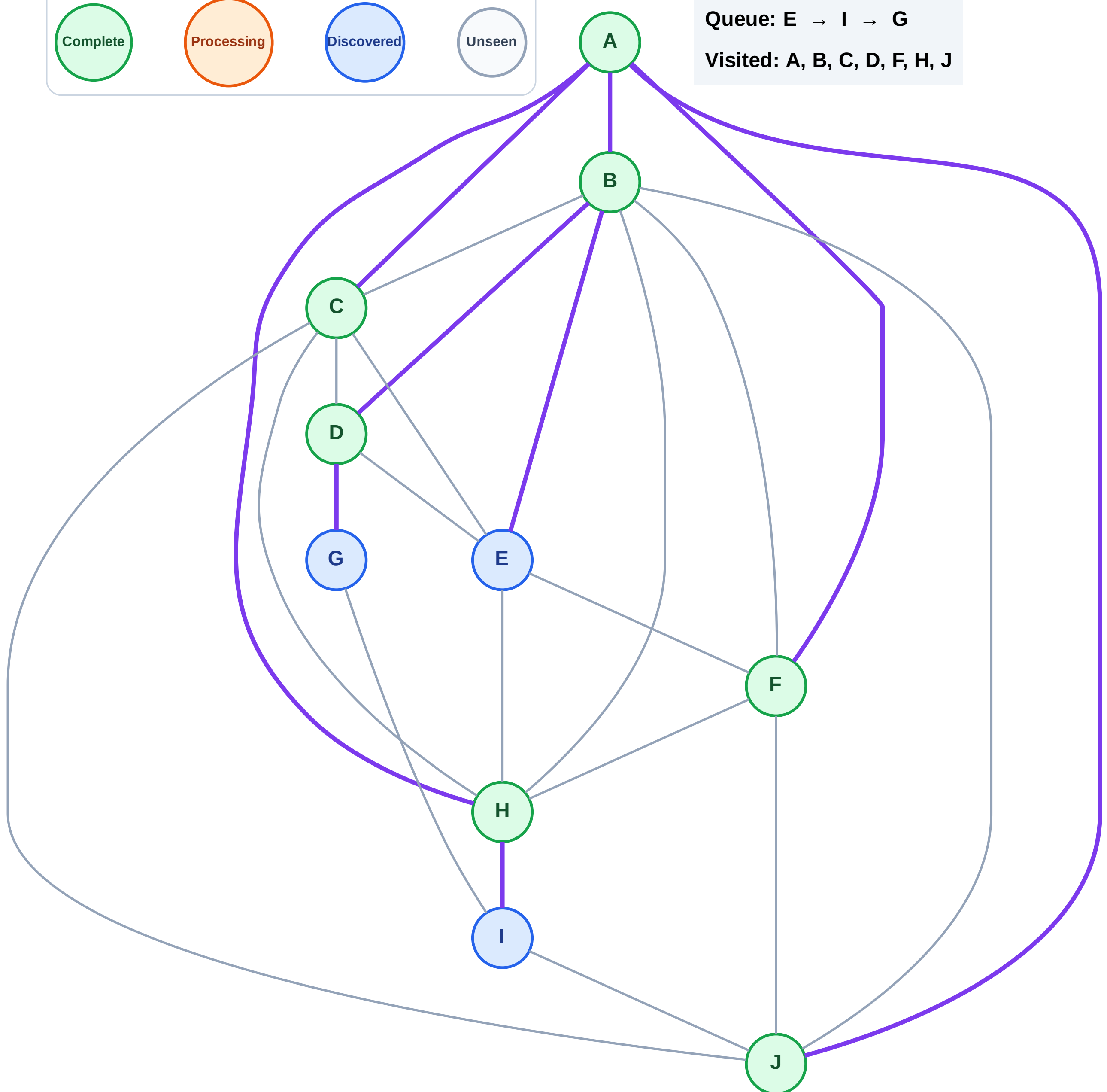
Discovered



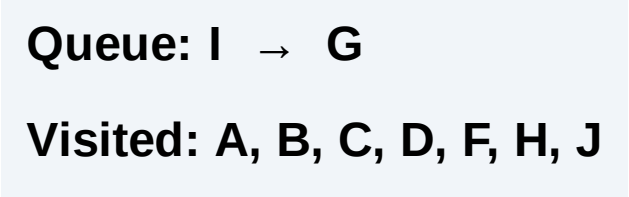
Unseen

Queue: E → I → G

Visited: A, B, C, D, F, H, J



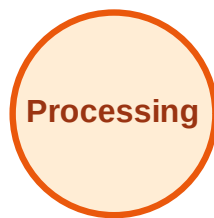
Step 16: Process node E



BFS Traversal

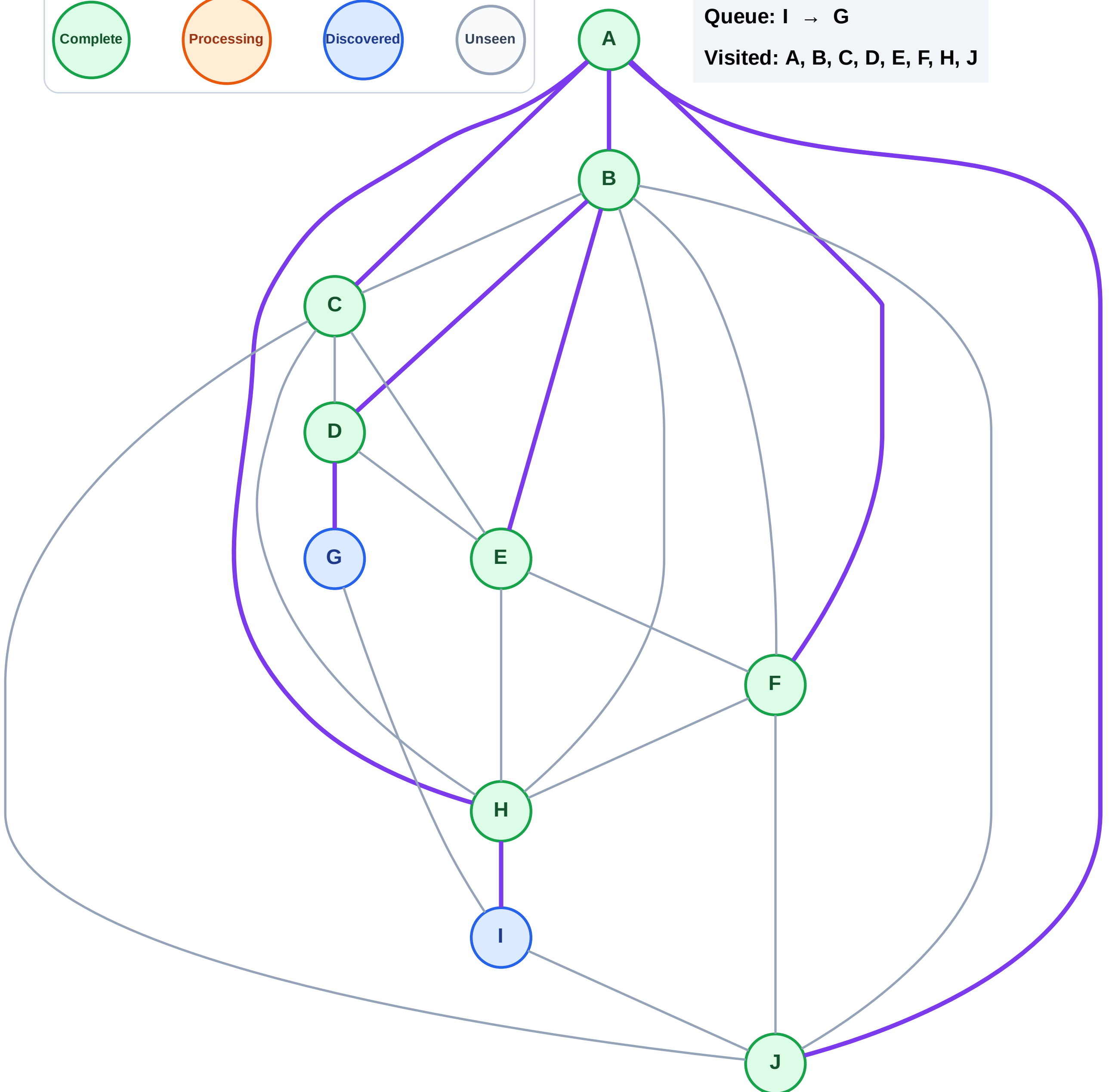
Step 17: No new neighbors from E

Legend



Queue: I → G

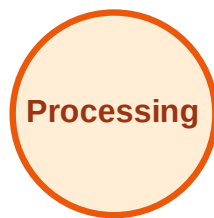
Visited: A, B, C, D, E, F, H, J



BFS Traversal

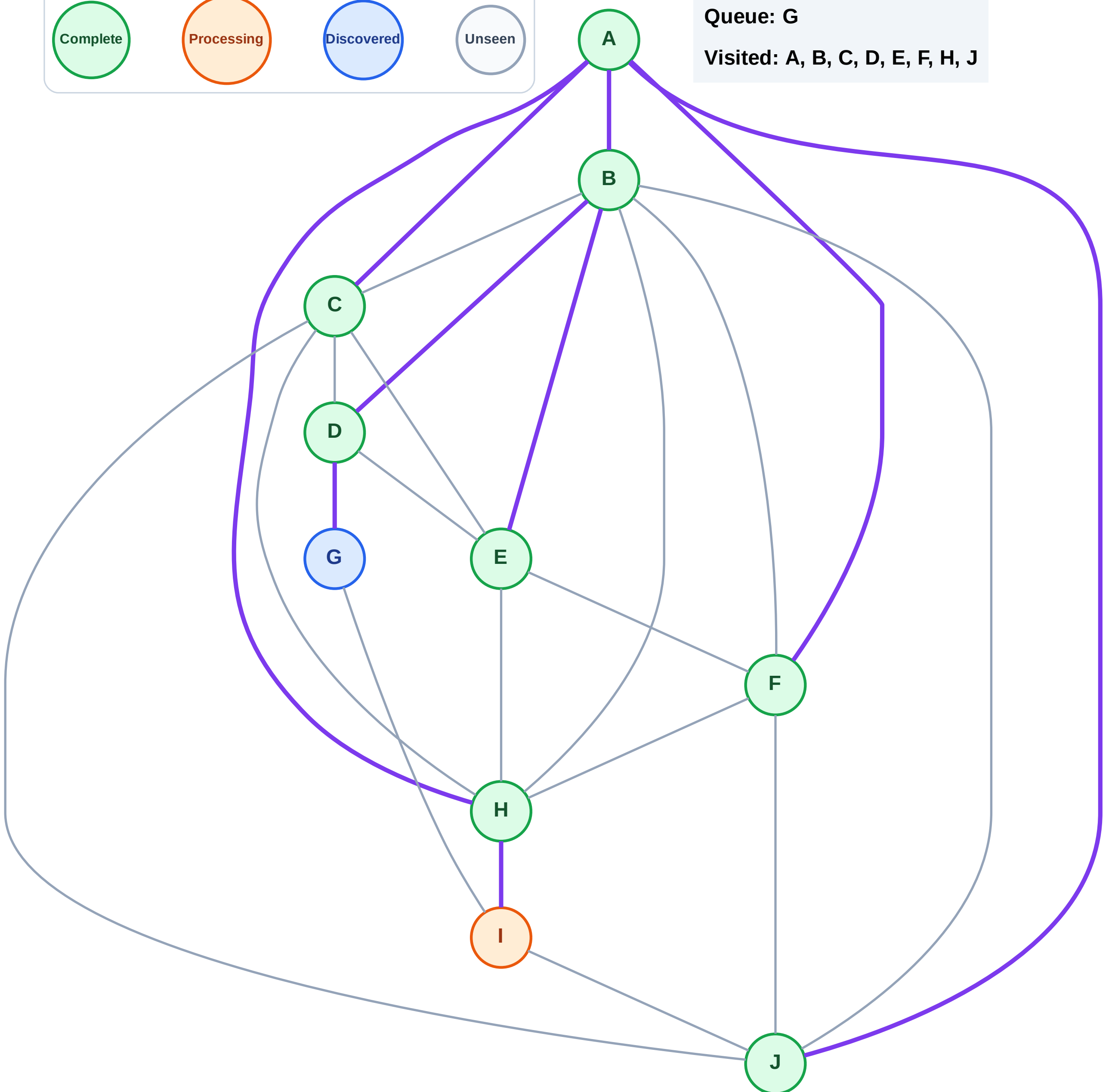
Step 18: Process node I

Legend



Queue: G

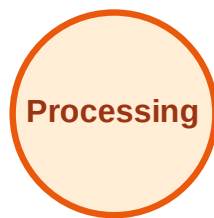
Visited: A, B, C, D, E, F, H, J



BFS Traversal

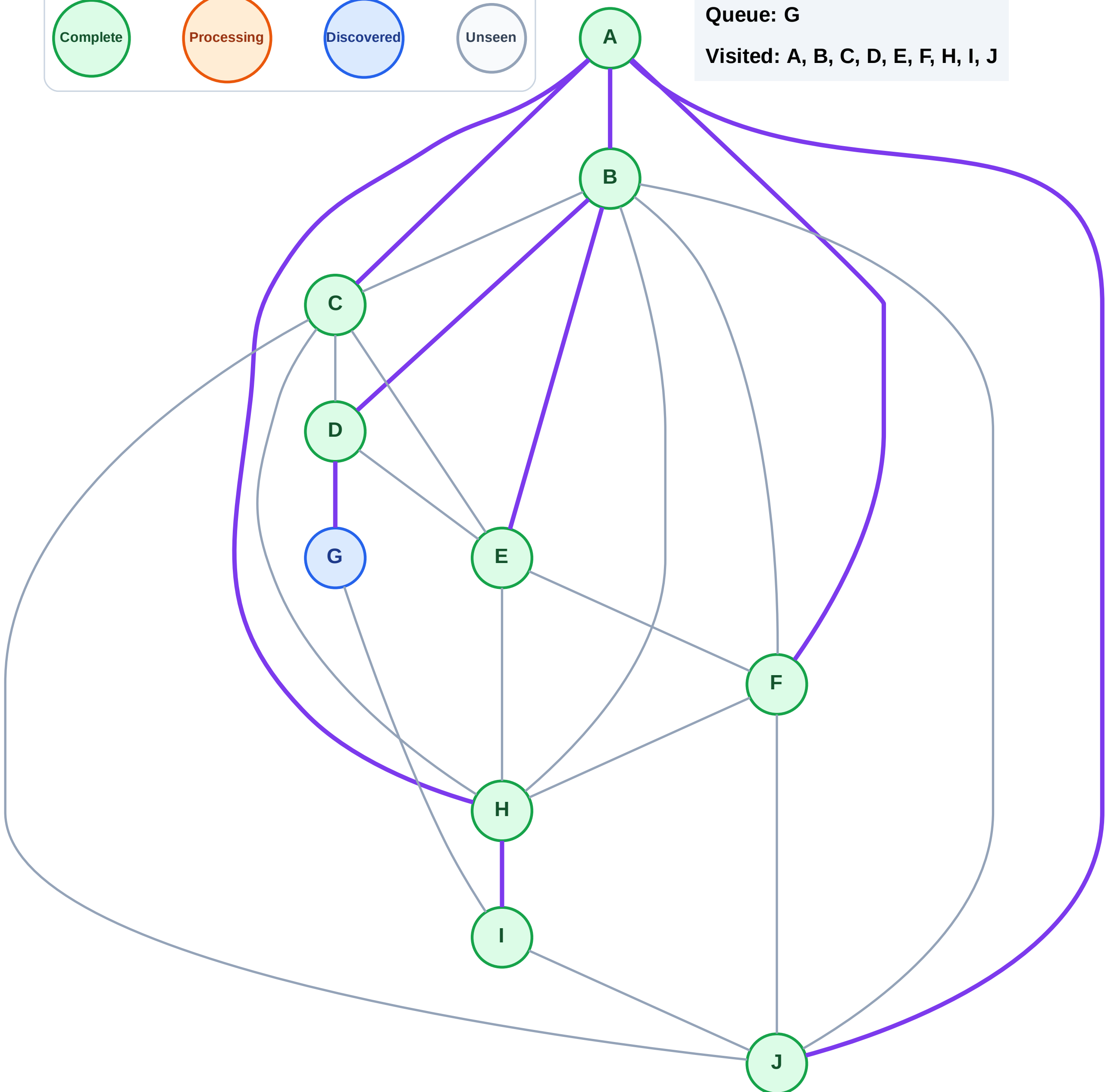
Step 19: No new neighbors from I

Legend

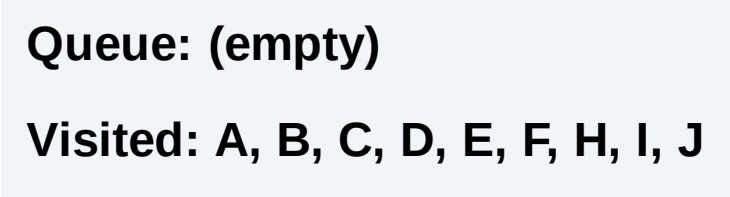


Queue: G

Visited: A, B, C, D, E, F, H, I, J



Step 20: Process node G



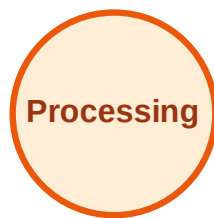
BFS Traversal

Step 21: No new neighbors from G

Legend



Complete



Processing



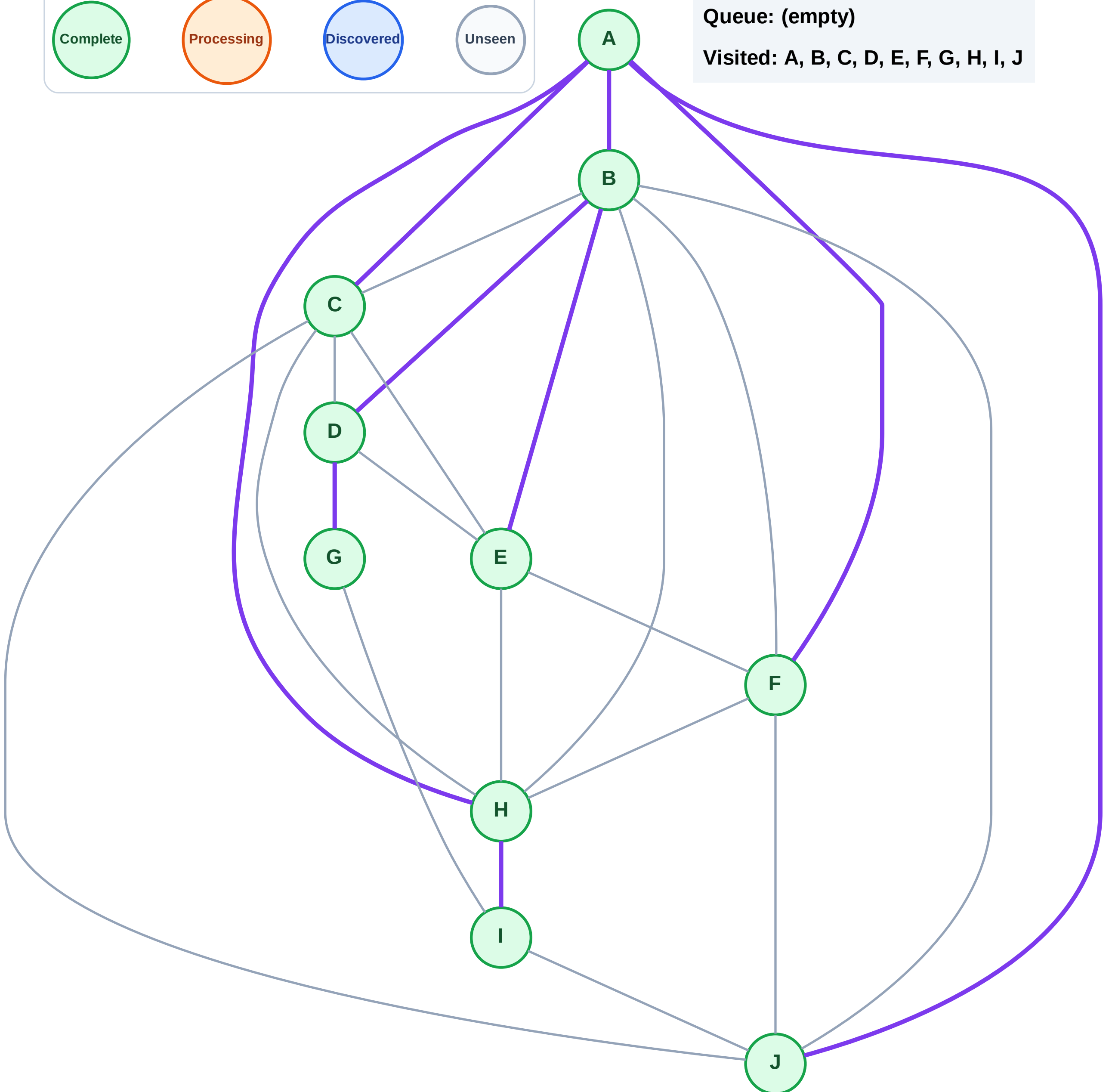
Discovered



Unseen

Queue: (empty)

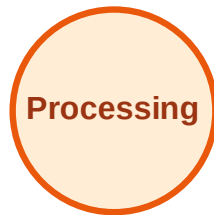
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

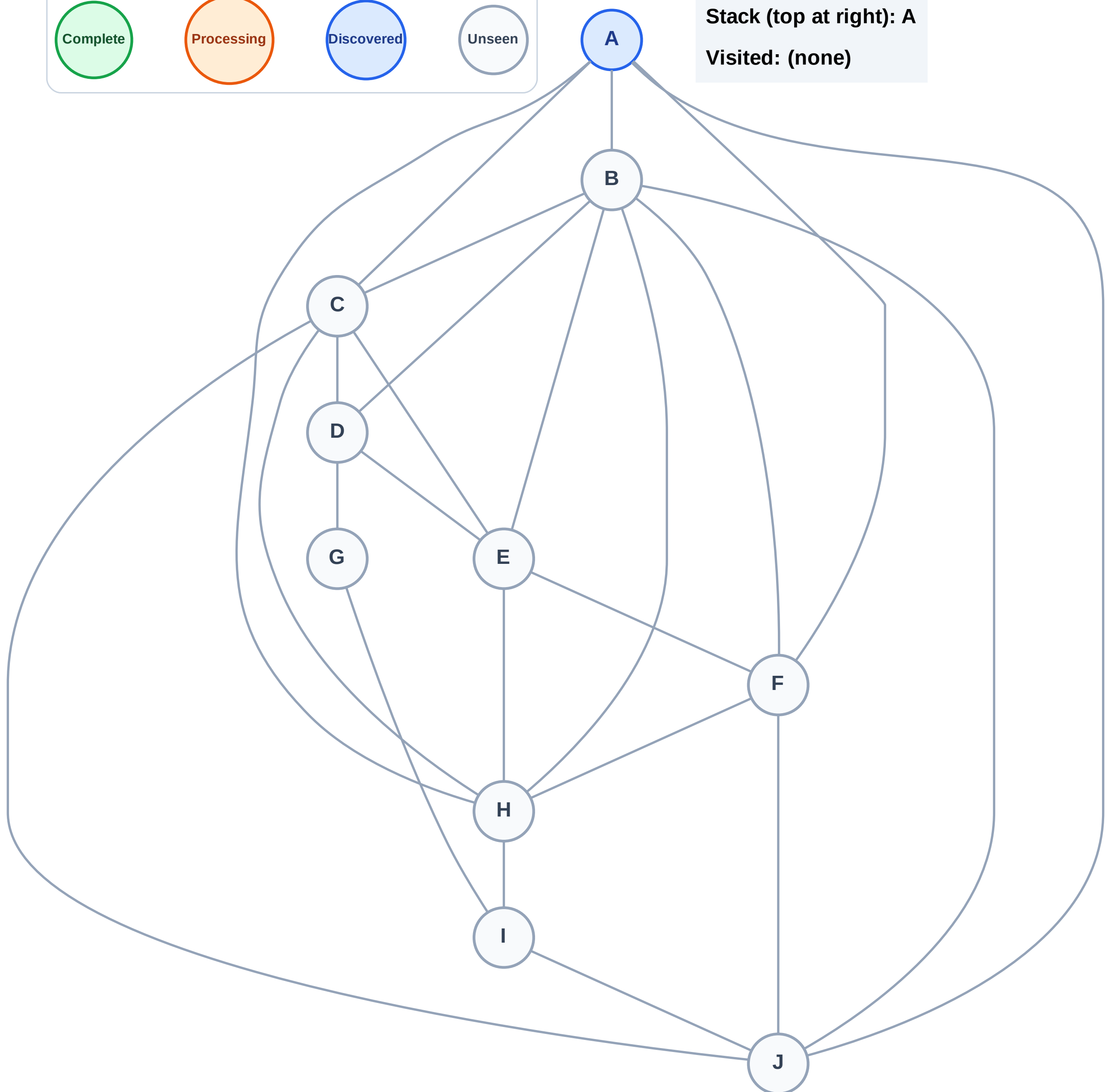
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



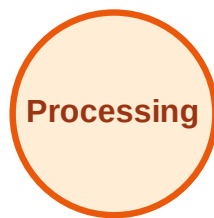
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



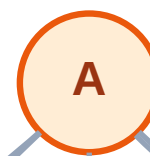
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



B

C

D

G

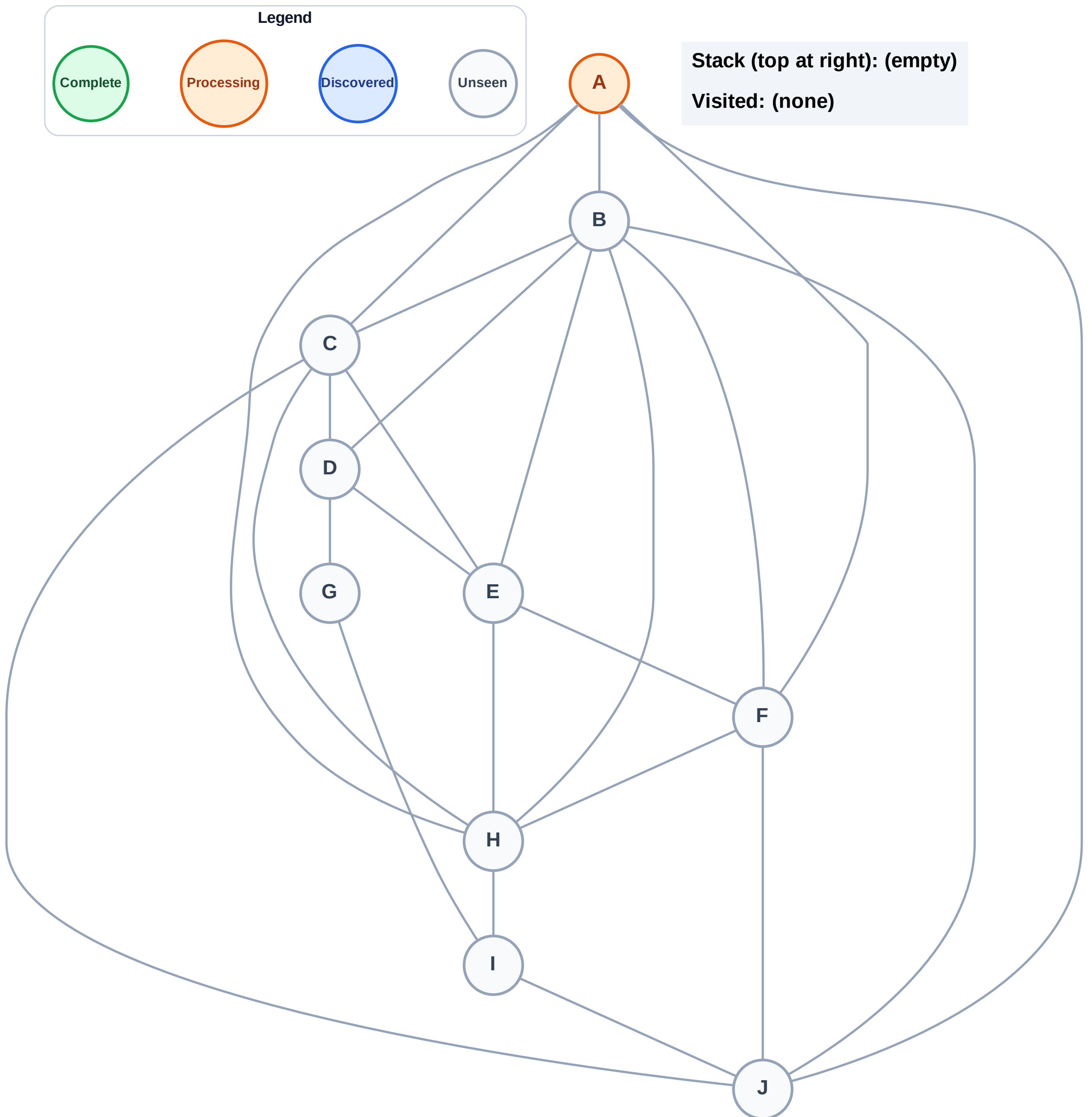
E

H

I

F

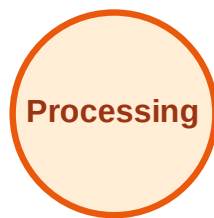
J



DFS Traversal

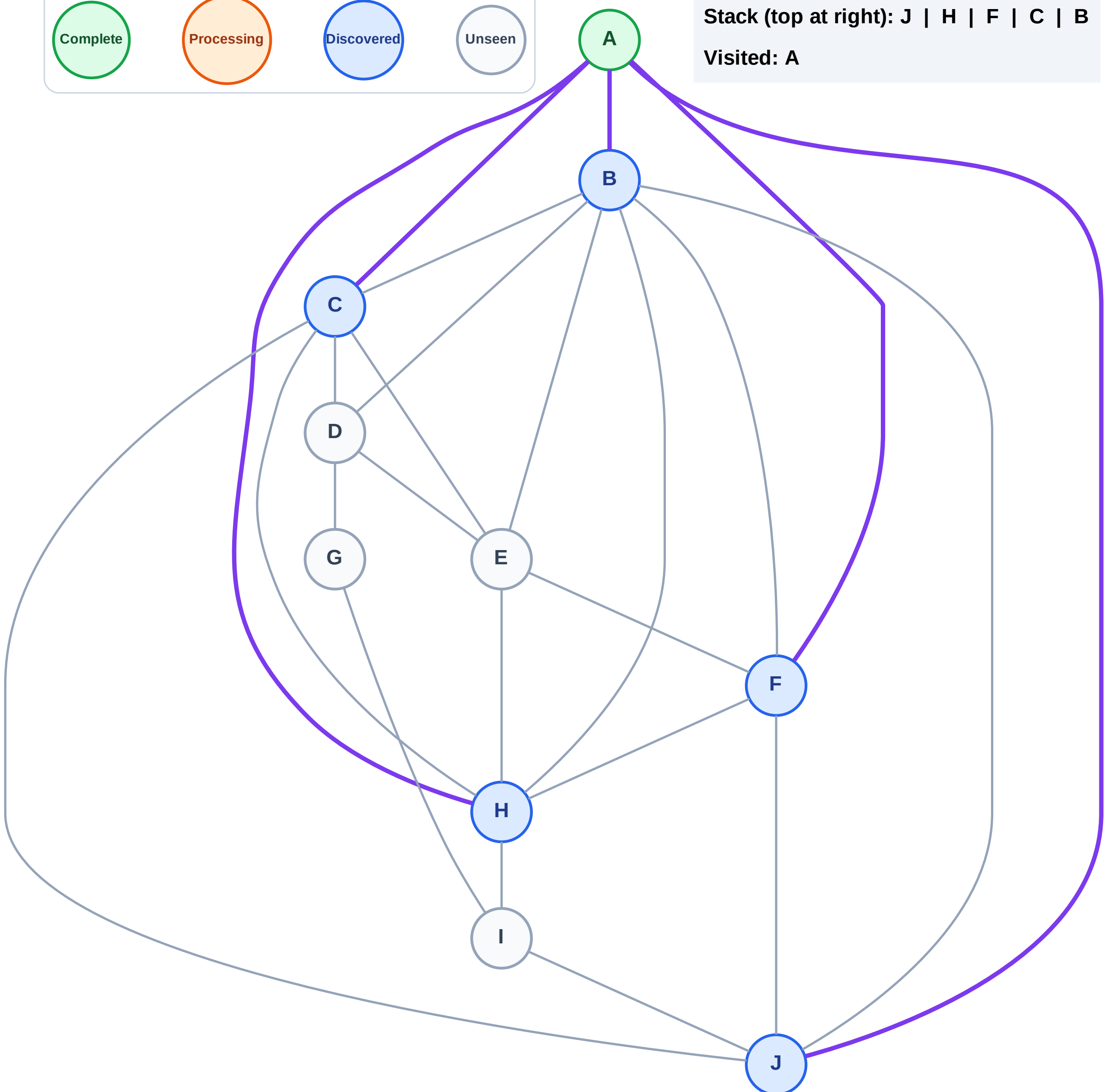
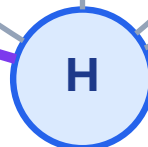
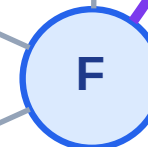
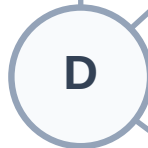
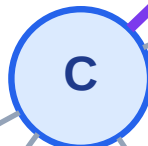
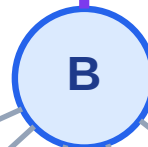
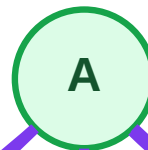
Step 3: Discover J, H, F, C, B from A

Legend



Stack (top at right): J | H | F | C | B

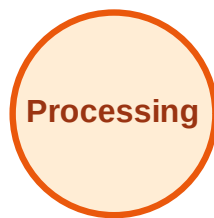
Visited: A



DFS Traversal

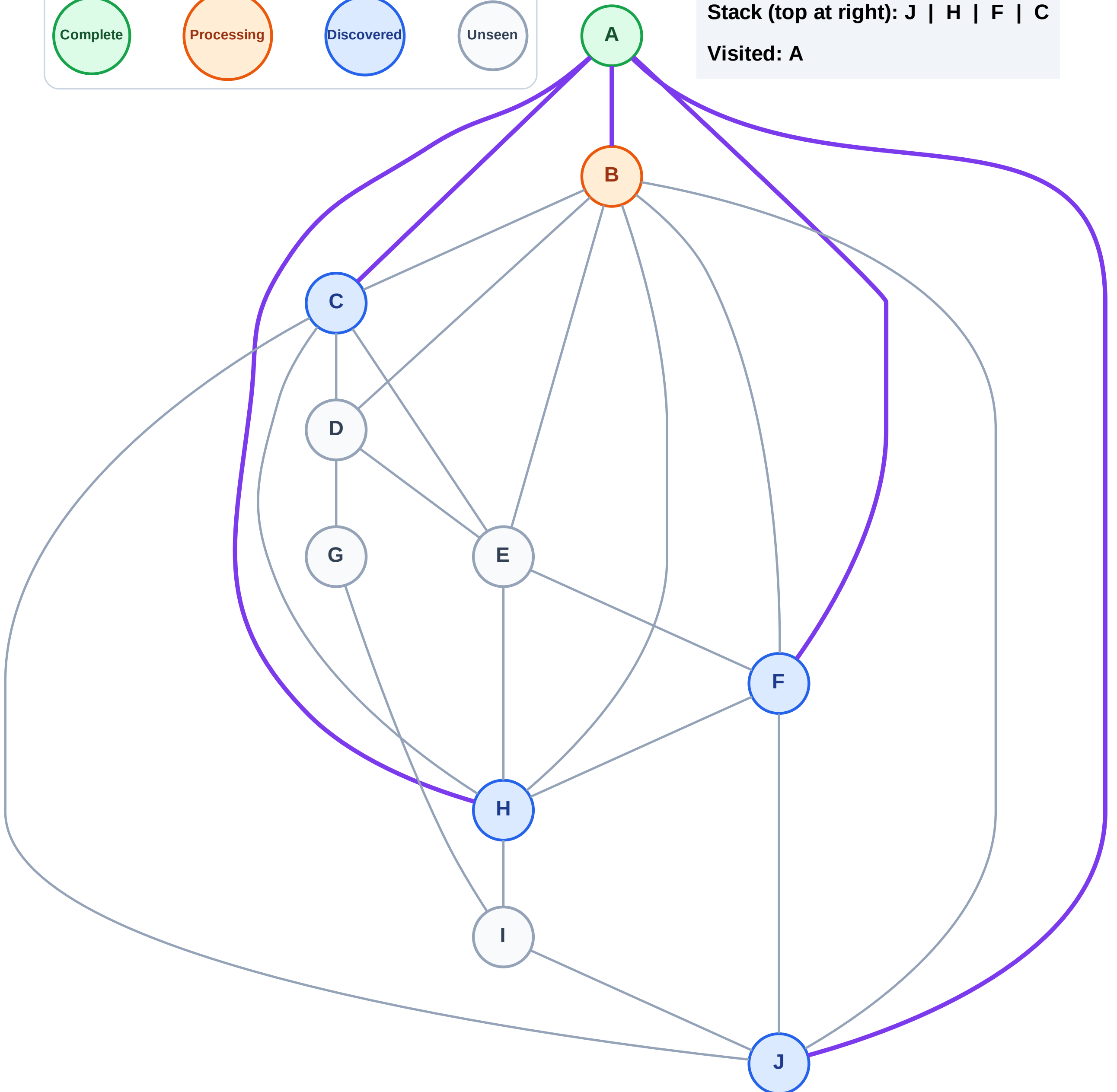
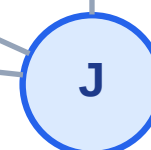
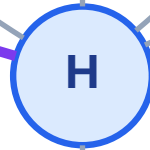
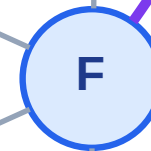
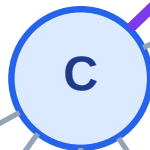
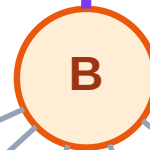
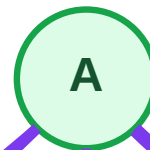
Step 4: Process node B

Legend



Stack (top at right): J | H | F | C

Visited: A



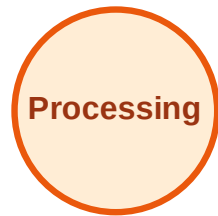
DFS Traversal

Step 5: Discover E, D from B

Legend



Complete



Processing



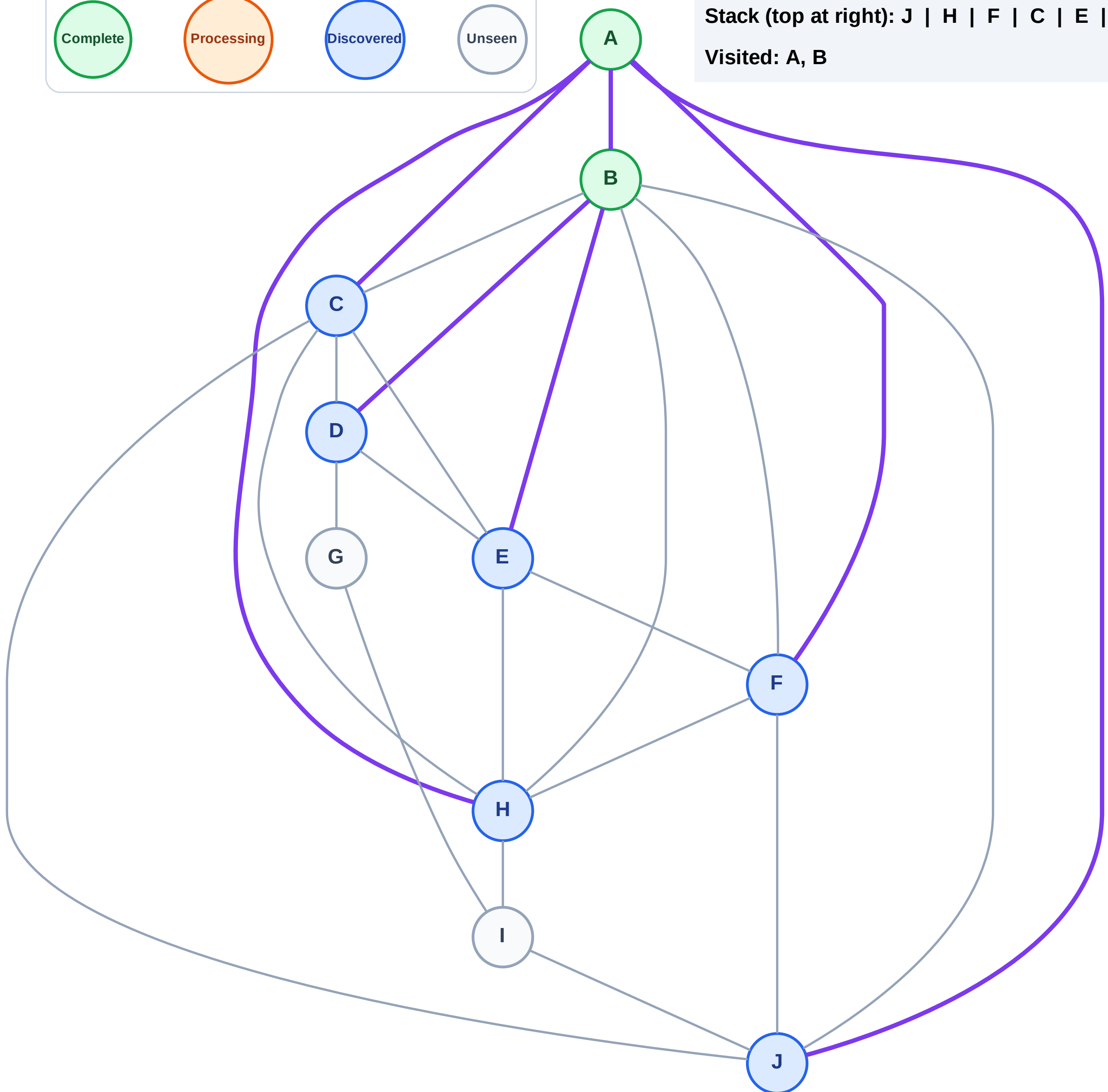
Discovered



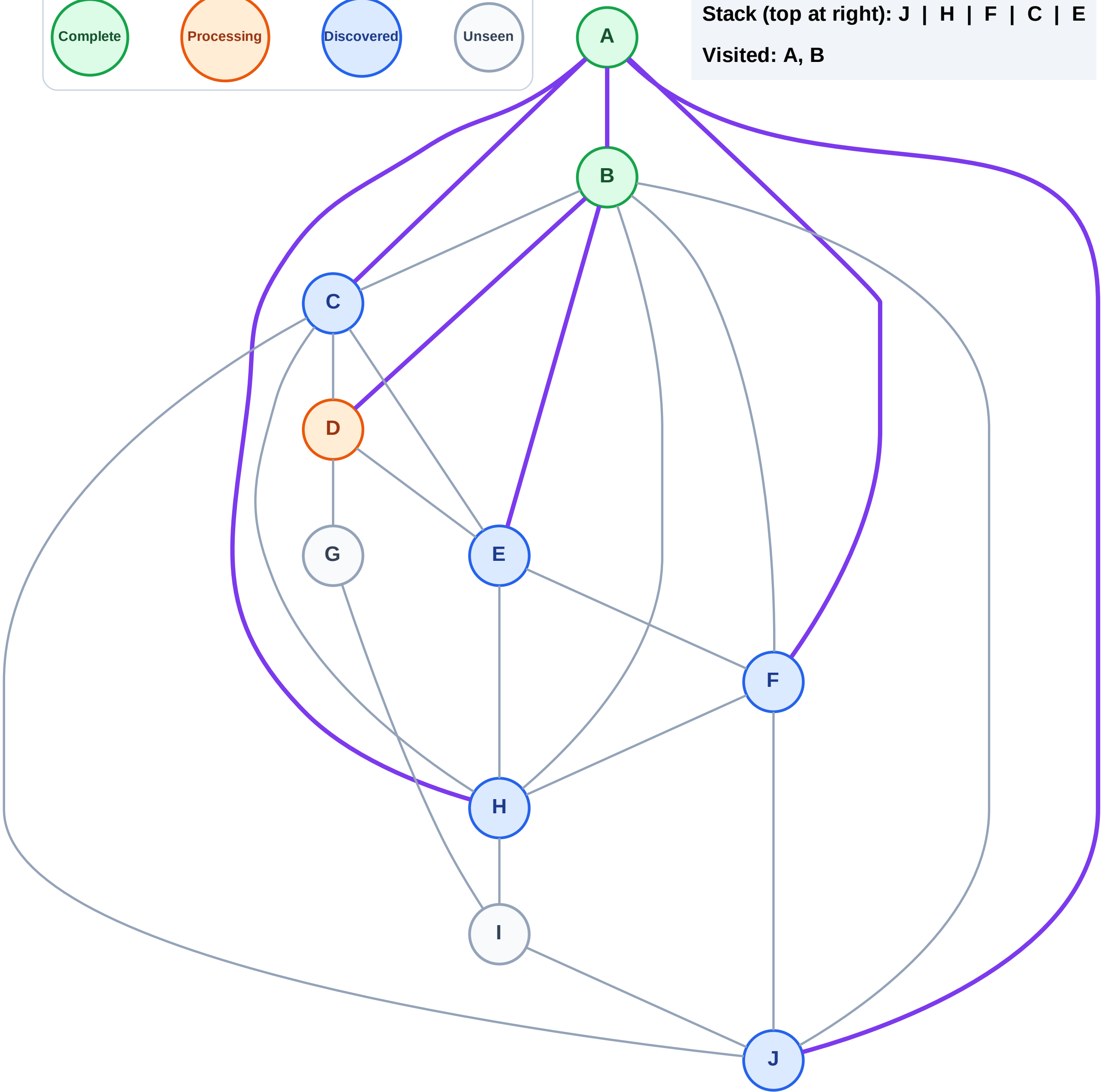
Unseen

Stack (top at right): J | H | F | C | E | D

Visited: A, B



Step 6: Process node D



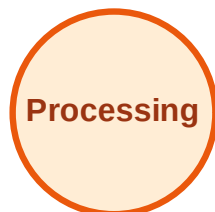
DFS Traversal

Step 7: Discover G from D

Legend



Complete



Processing



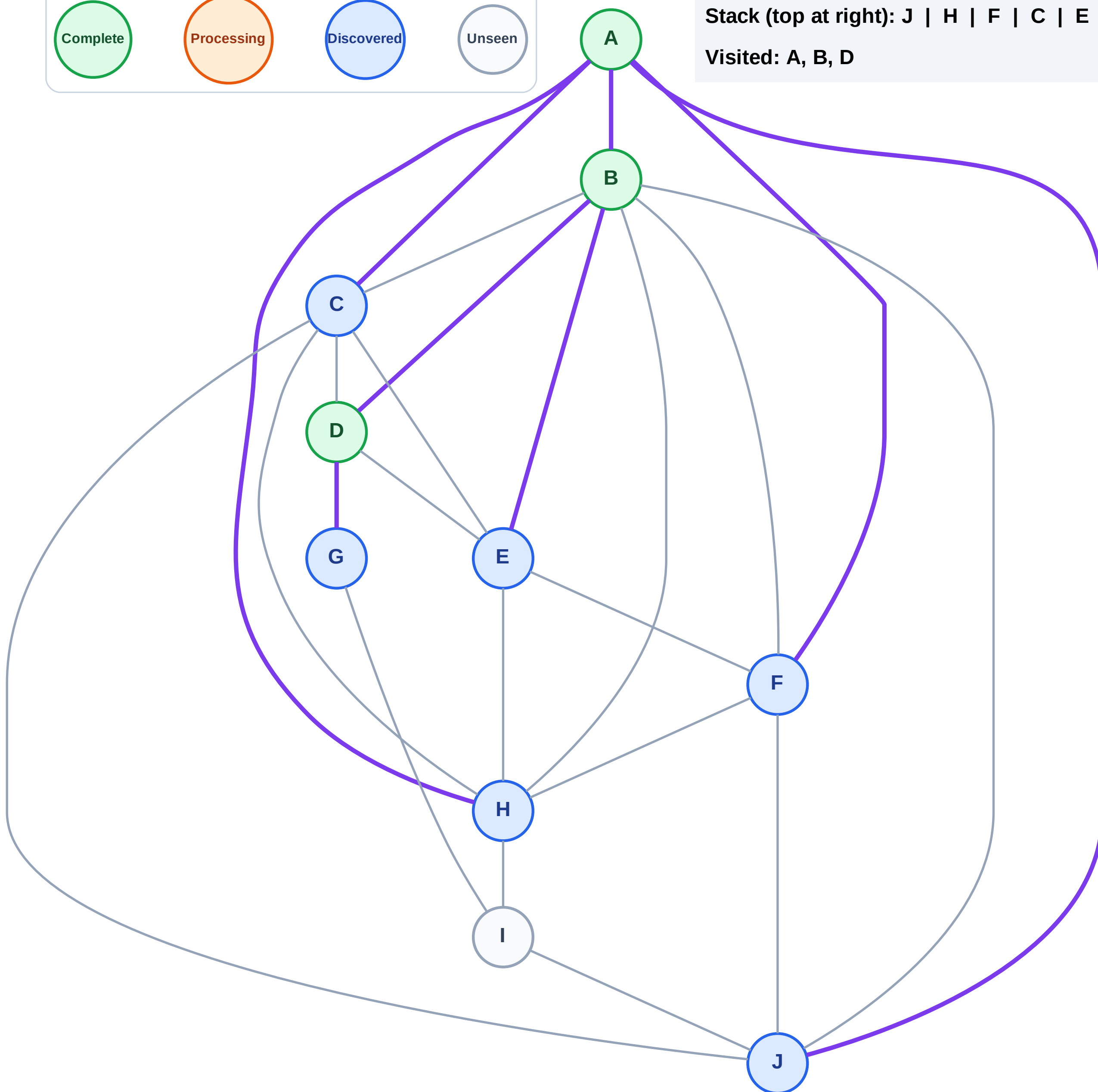
Discovered



Unseen

Stack (top at right): J | H | F | C | E | G

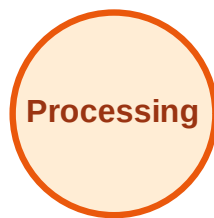
Visited: A, B, D



DFS Traversal

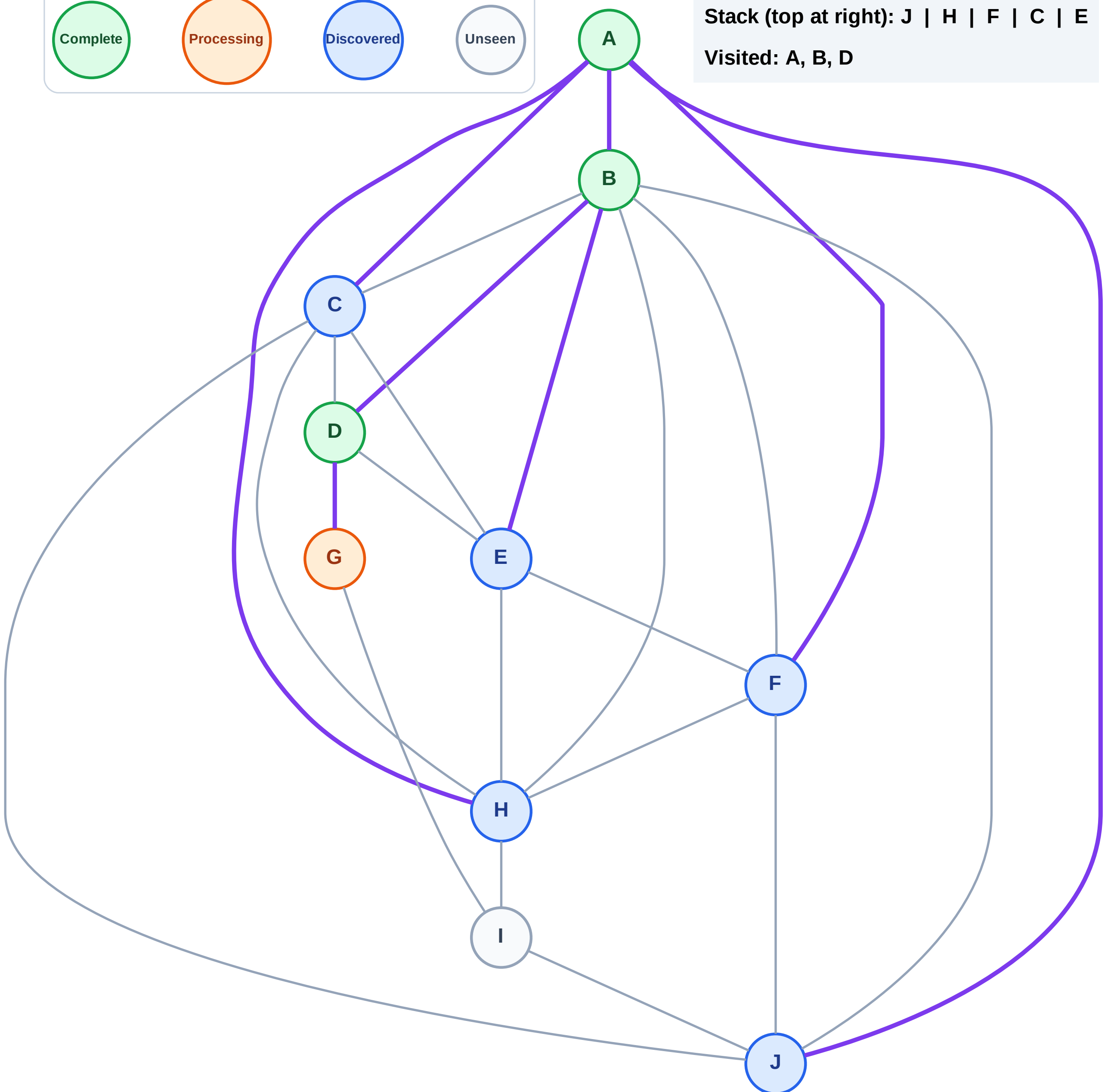
Step 8: Process node G

Legend



Stack (top at right): J | H | F | C | E

Visited: A, B, D



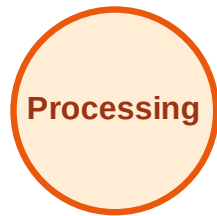
DFS Traversal

Step 9: Discover I from G

Legend



Complete



Processing



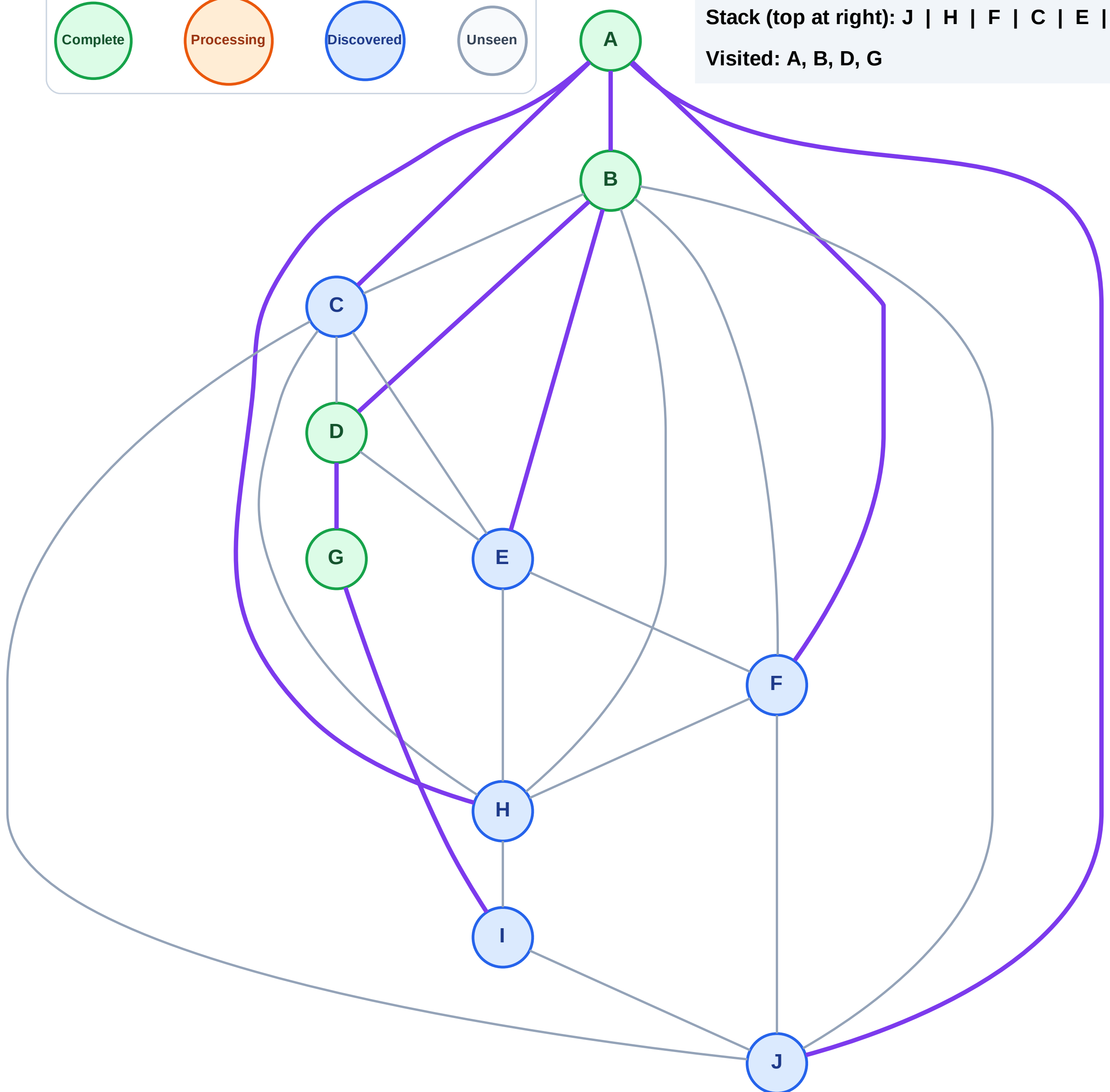
Discovered



Unseen

Stack (top at right): J | H | F | C | E | I

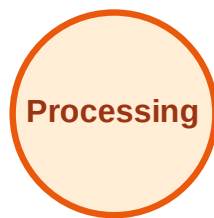
Visited: A, B, D, G



DFS Traversal

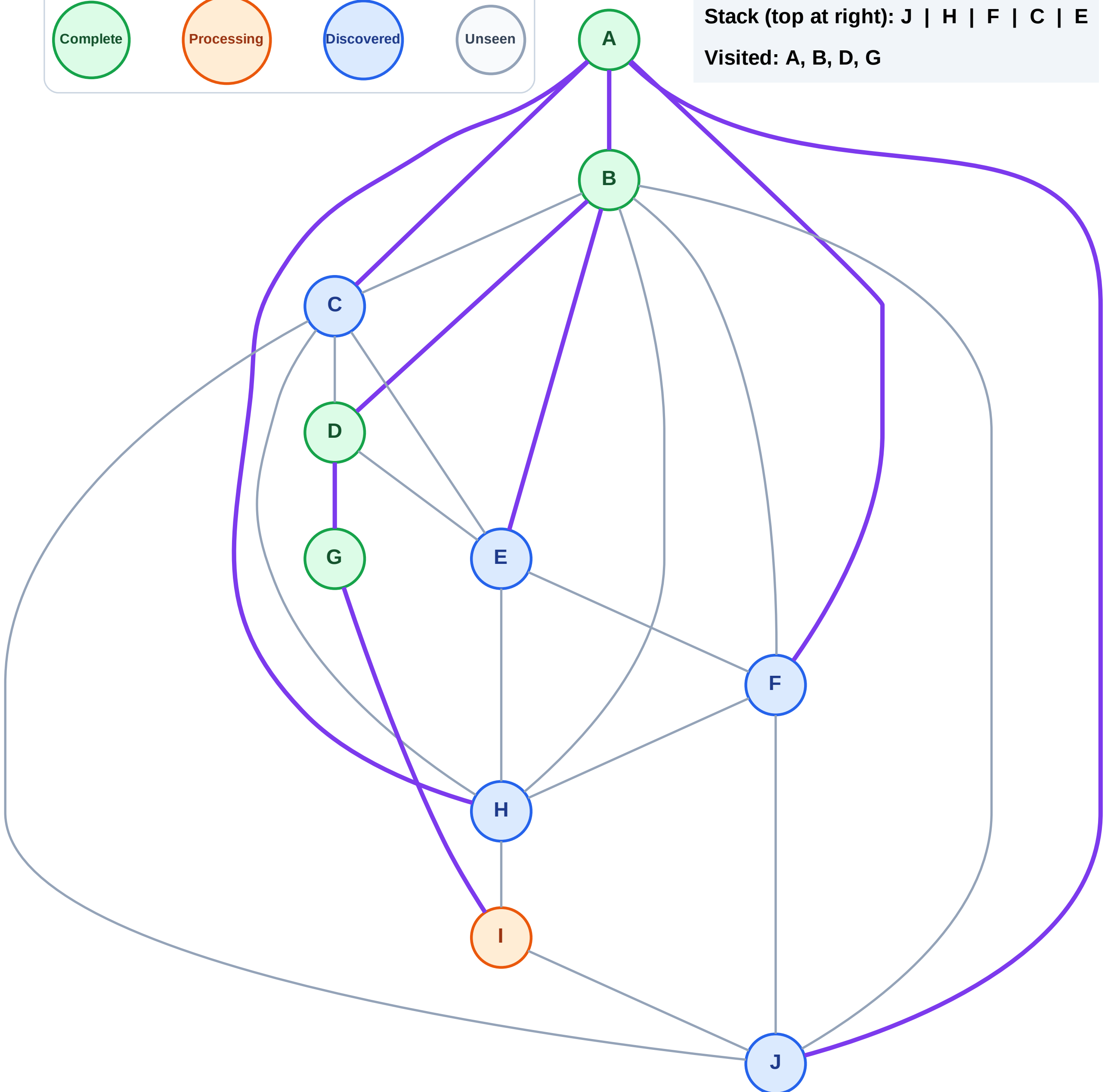
Step 10: Process node I

Legend



Stack (top at right): J | H | F | C | E

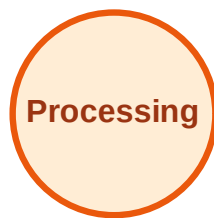
Visited: A, B, D, G



DFS Traversal

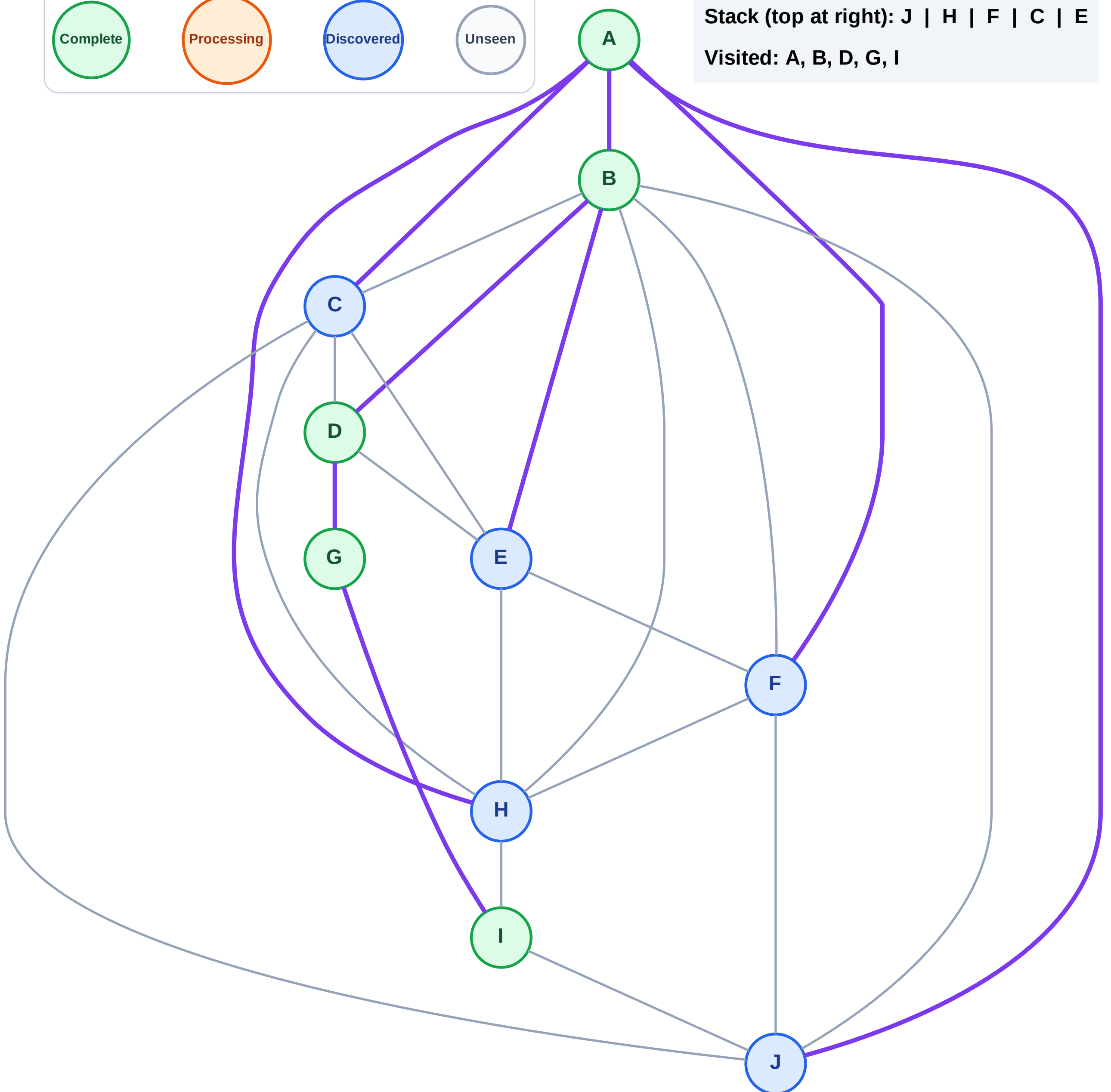
Step 11: No new neighbors from I

Legend



Stack (top at right): J | H | F | C | E

Visited: A, B, D, G, I



DFS Traversal

Step 12: Process node E

Legend

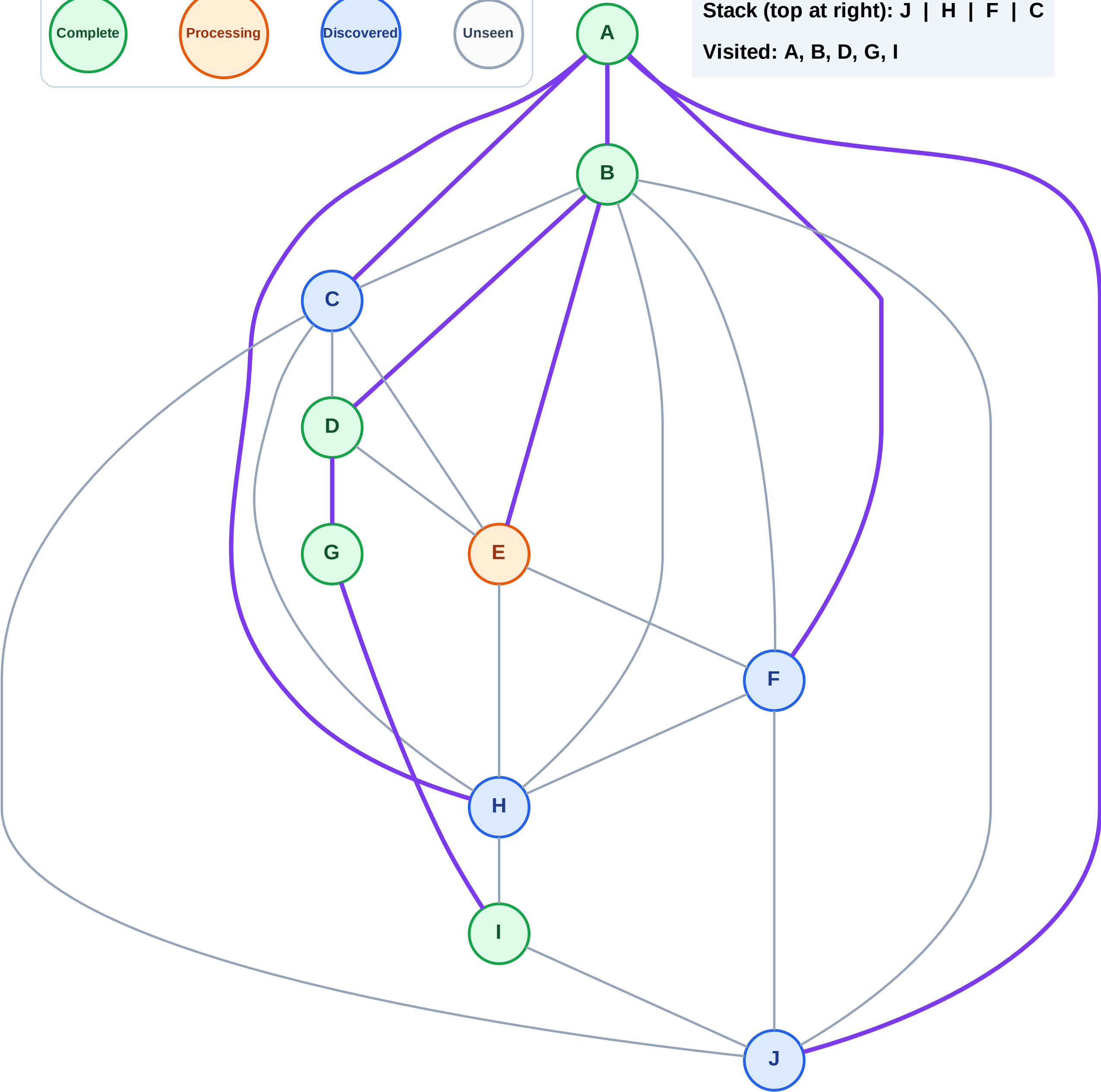
Complete

Processing

Discovered

Unseen

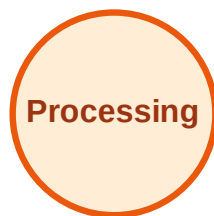
Stack (top at right): J | H | F | C
Visited: A, B, D, G, I



DFS Traversal

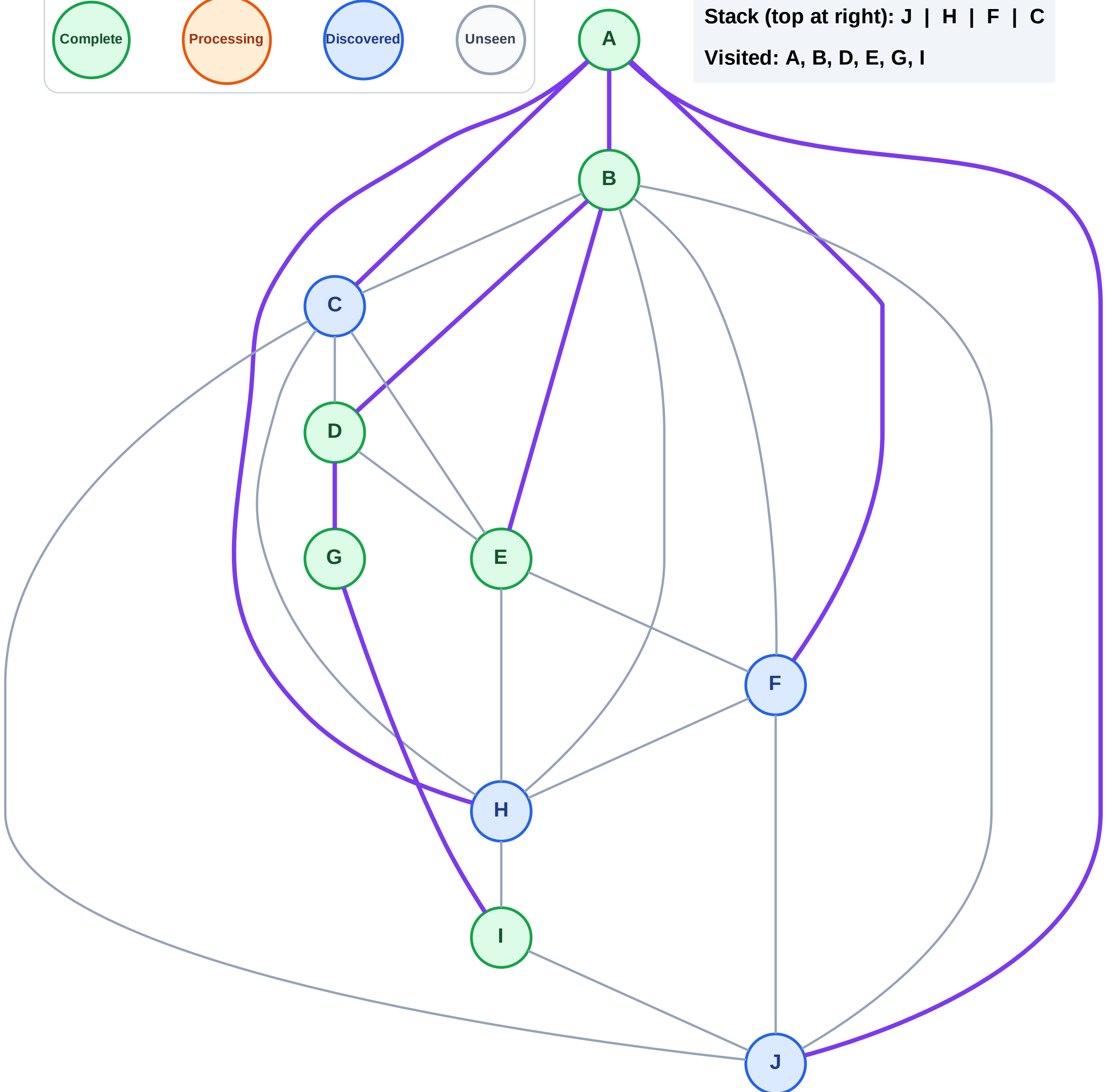
Step 13: No new neighbors from E

Legend



Stack (top at right): J | H | F | C

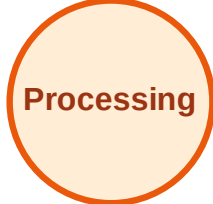
Visited: A, B, D, E, G, I



DFS Traversal

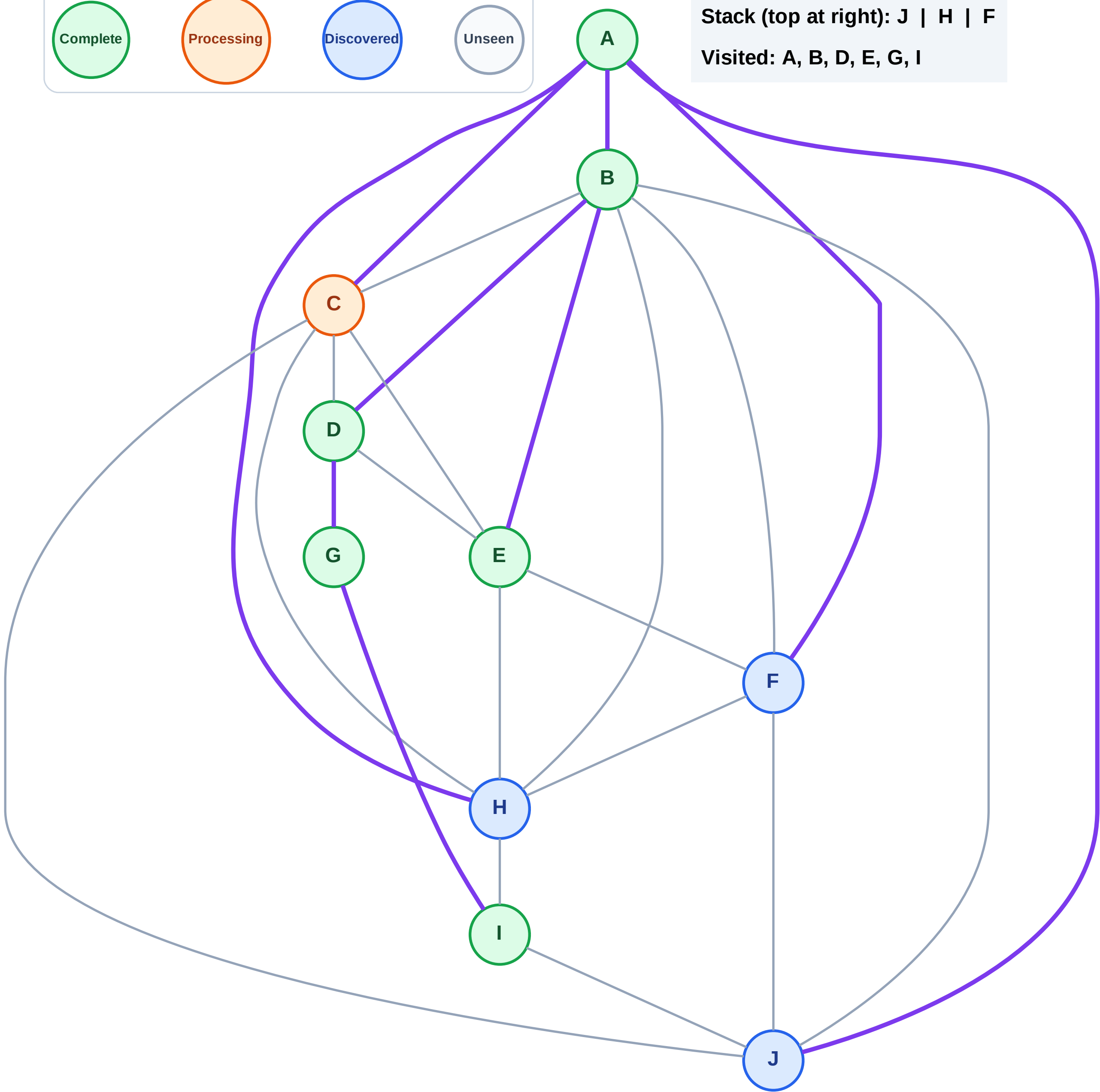
Step 14: Process node C

Legend



Stack (top at right): J | H | F

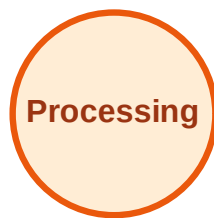
Visited: A, B, D, E, G, I



DFS Traversal

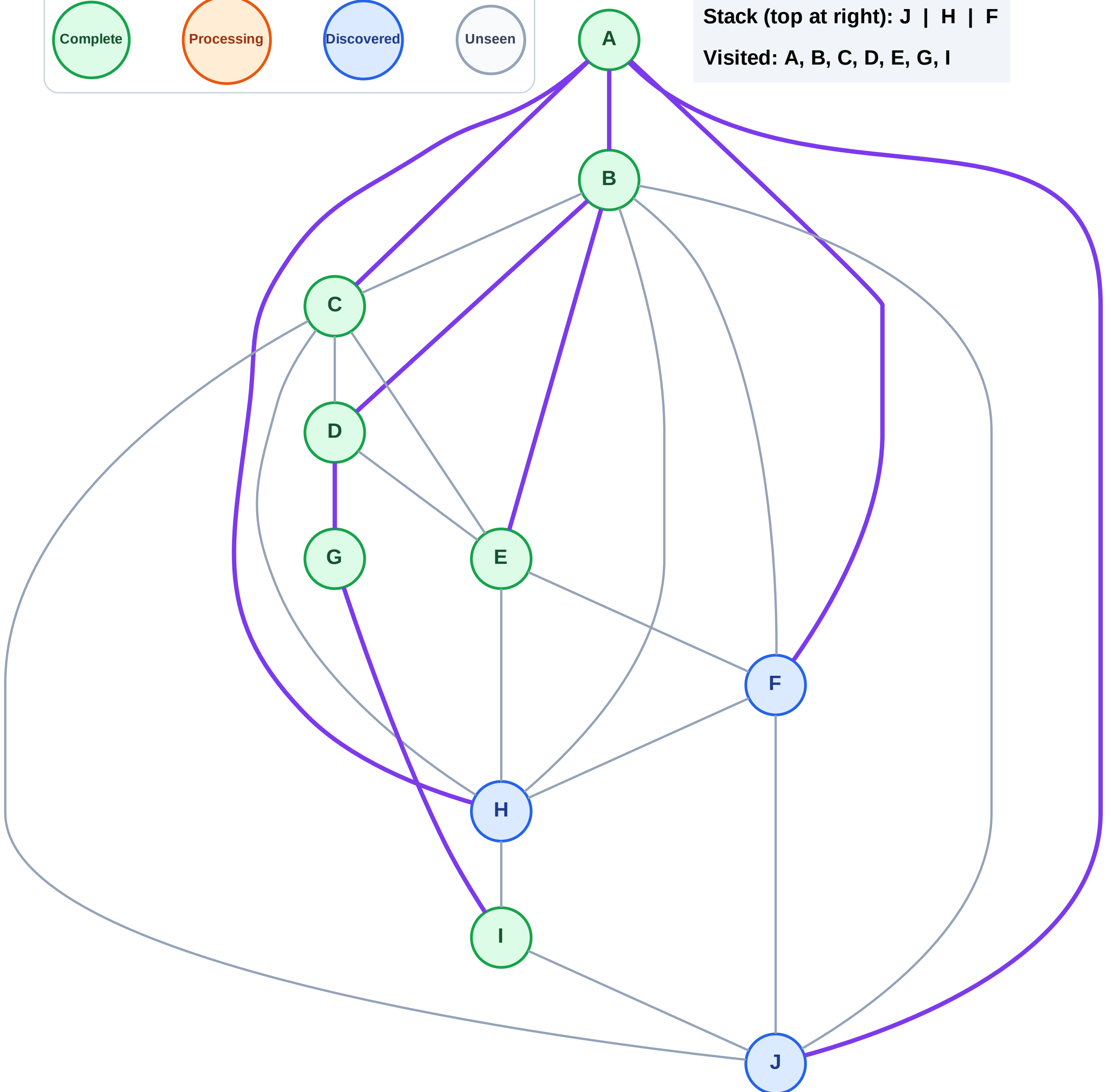
Step 15: No new neighbors from C

Legend



Stack (top at right): J | H | F

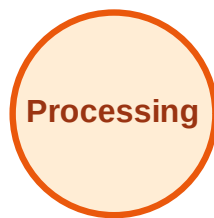
Visited: A, B, C, D, E, G, I



DFS Traversal

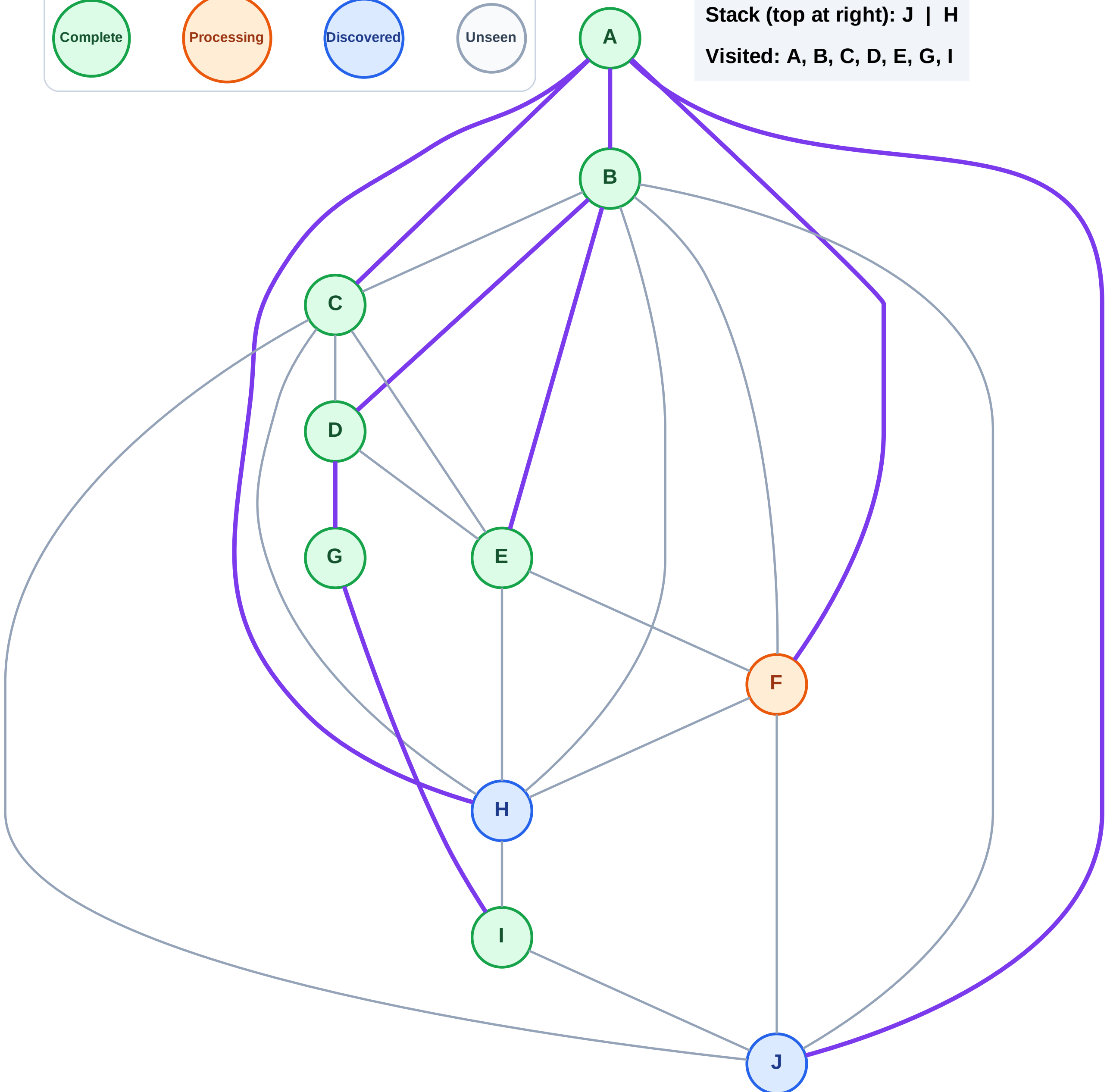
Step 16: Process node F

Legend



Stack (top at right): J | H

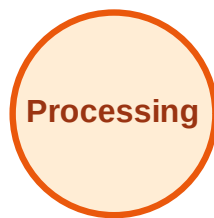
Visited: A, B, C, D, E, G, I



DFS Traversal

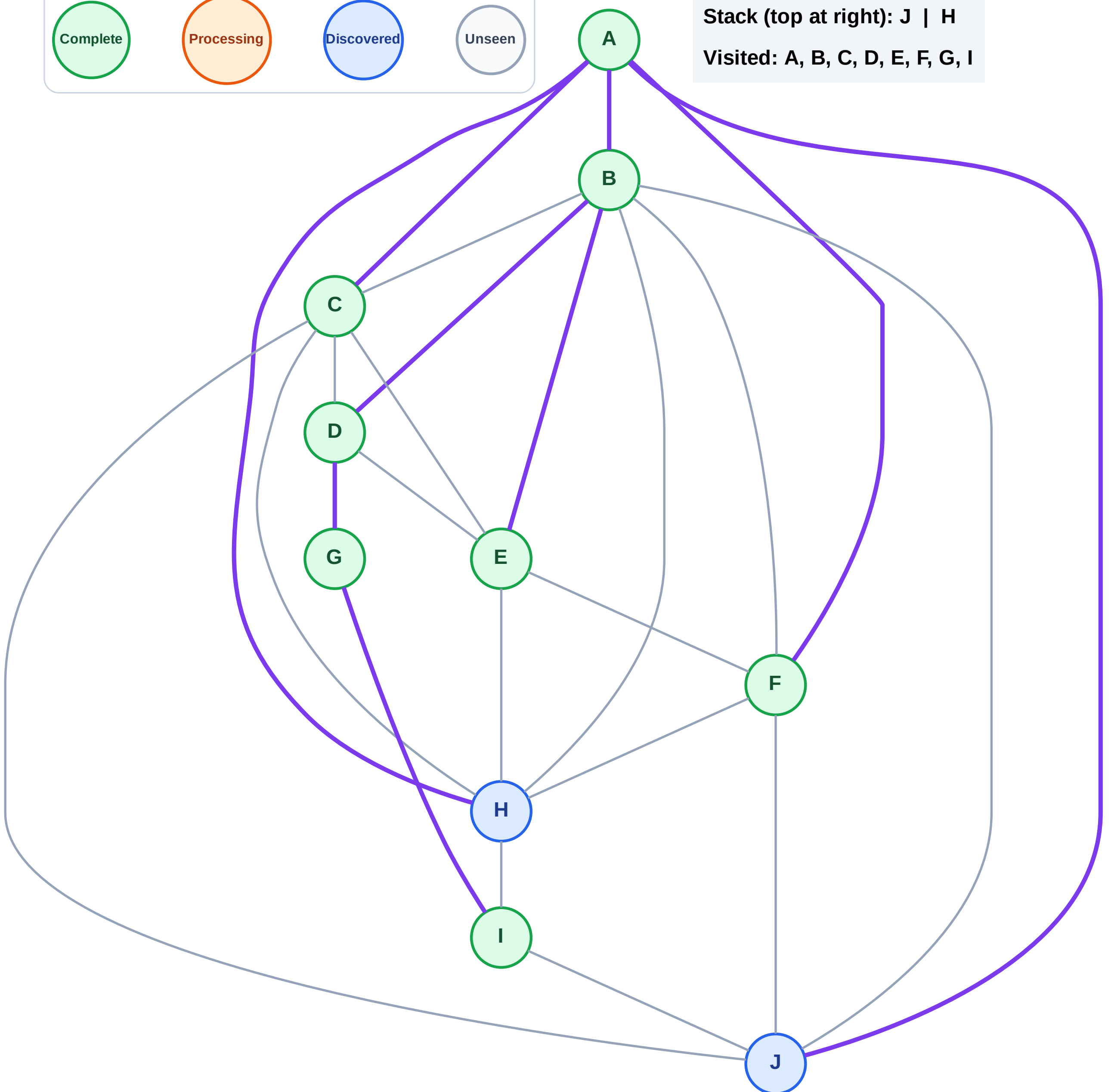
Step 17: No new neighbors from F

Legend



Stack (top at right): J | H

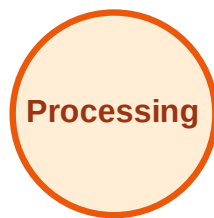
Visited: A, B, C, D, E, F, G, I



DFS Traversal

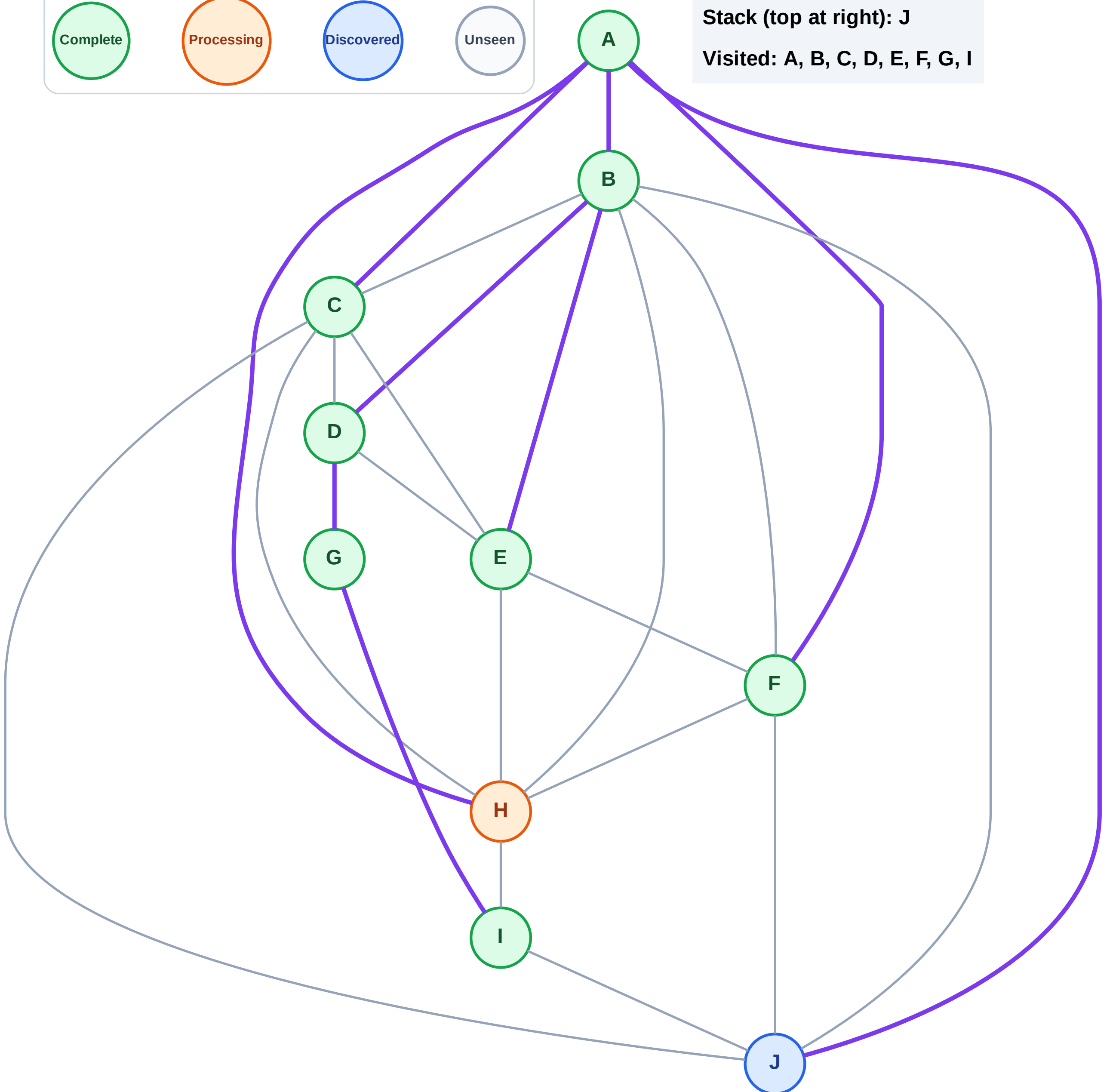
Step 18: Process node H

Legend



Stack (top at right): J

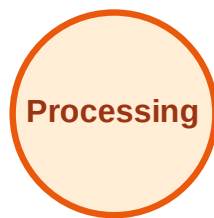
Visited: A, B, C, D, E, F, G, I



DFS Traversal

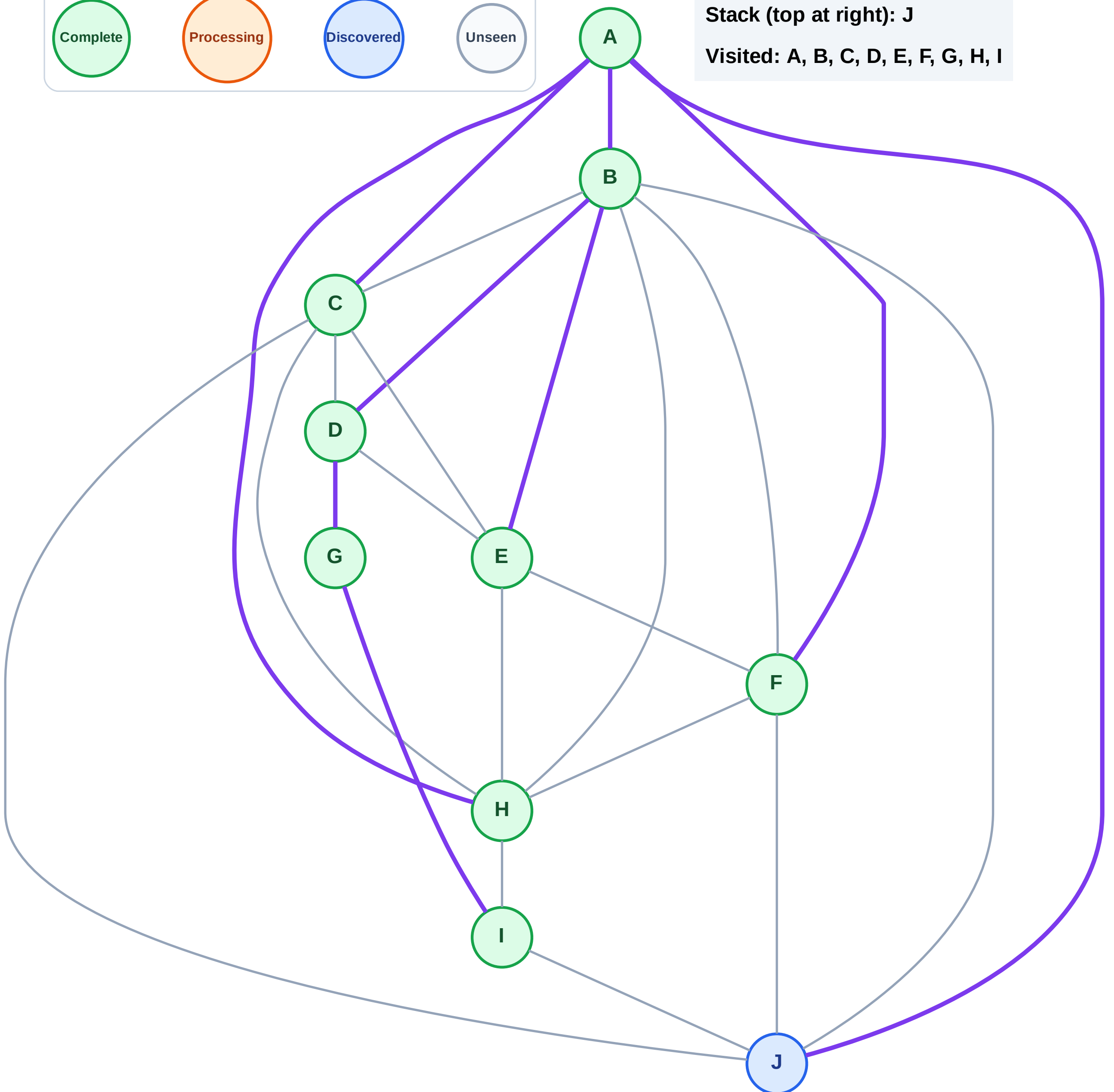
Step 19: No new neighbors from H

Legend



Stack (top at right): J

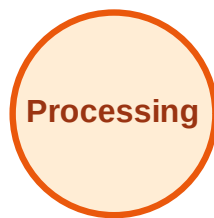
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

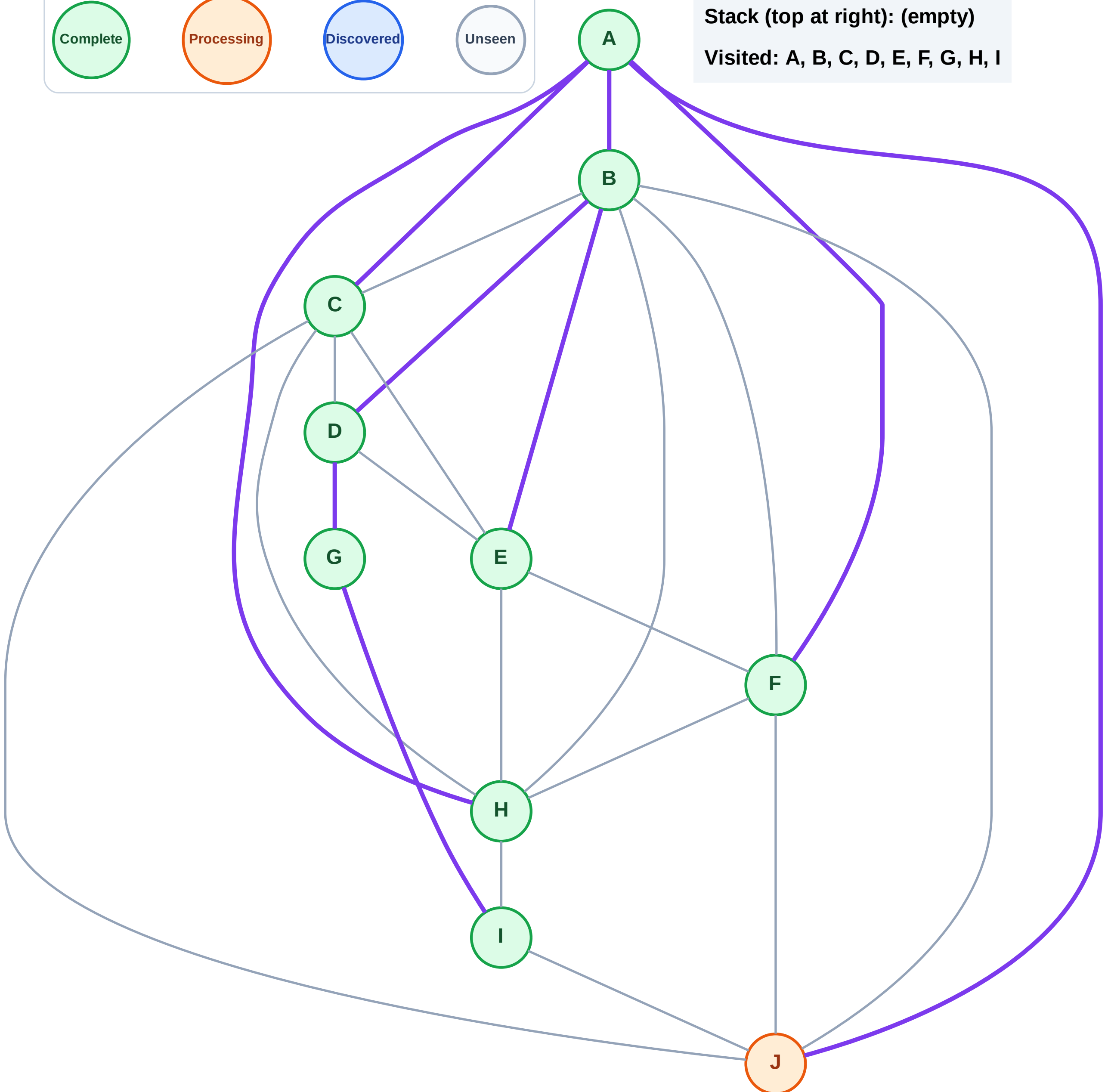
Step 20: Process node J

Legend



Stack (top at right): (empty)

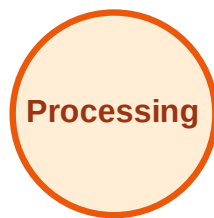
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

